

Modern Networking

Unit 1 – Modern Networking

Chapter 1 : Elements of Modern Networking

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Unit	Details
I	Modern Networking Elements of Modern Networking The Networking Ecosystem ,Example Network Architectures,Global Network Architecture,A Typical Network Hierarchy Ethernet Applications of Ethernet Standards Ethernet Data Rates Wi-Fi Applications of Wi-Fi,Standards Wi-Fi Data Rates 4G/5G Cellular First Generation Second Generation, Third Generation Fourth Generation Fifth Generation, Cloud Computing Cloud Computing Concepts The Benefits of Cloud Computing Cloud Networking Cloud Storage, Internet of Things Things on the Internet of Things, Evolution Layers of the Internet of Things, Network Convergence Unified Communications, Requirements and Technology Types of Network and Internet Traffic,Elastic Traffic,Inelastic Traffic, Real-Time Traffic Characteristics Demand: Big Data, Cloud Computing, and Mobile TrafficBig Data Cloud Computing,,Mobile Traffic, Requirements: QoS and QoE,,Quality of Service,Quality of Experience, Routing Characteristics, Packet Forwarding, Congestion Control ,Effects of Congestion,Congestion Control Techniques, SDN and NFV Software-Defined Networking,Network Functions Virtualization Modern Networking Elements

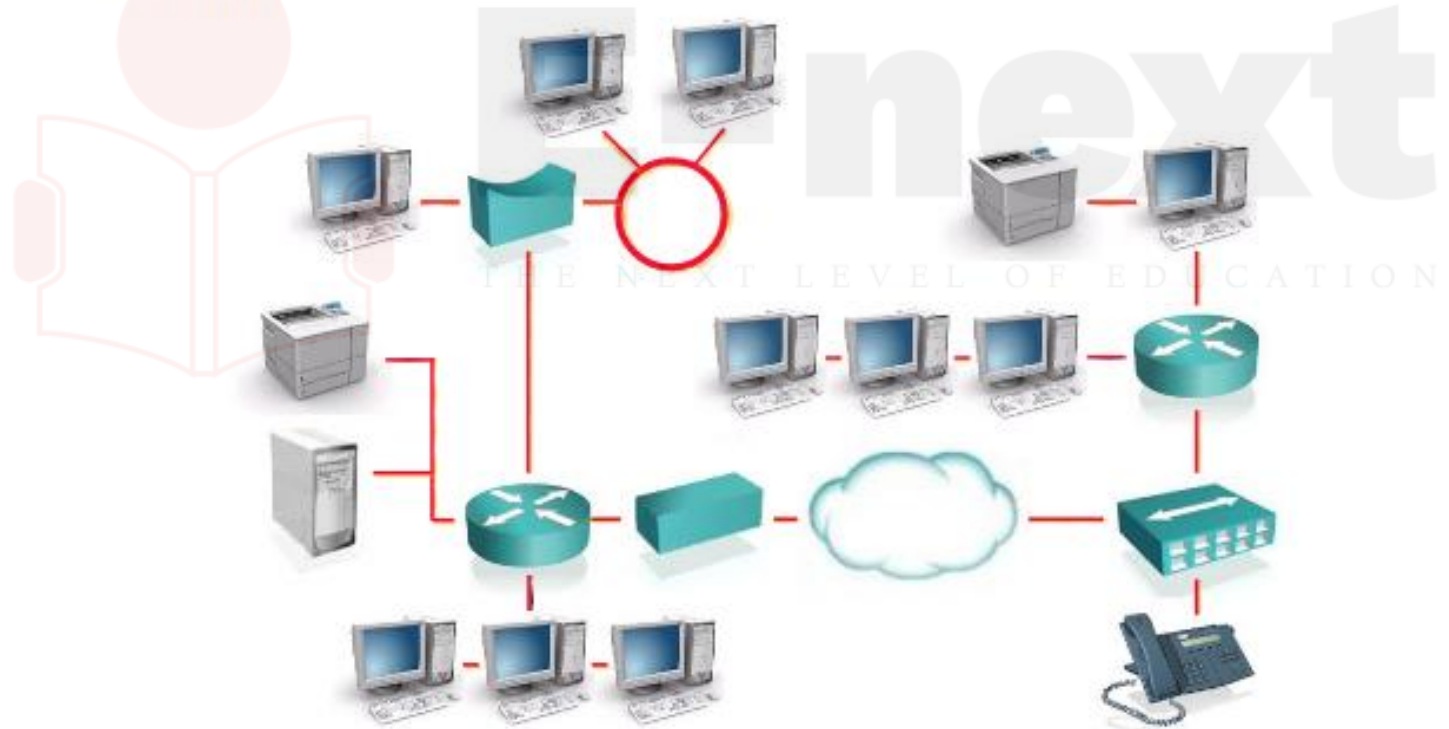
Semester – II		
Course Code	Course Title	Credits
PSIT201	Big Data Analytics	4
PSIT202	Modern Networking	4
PSIT203	Microservices Architecture	4
PSIT204	Image Processing	4
PSIT2P1	Big Data Analytics Practical	2
PSIT2P2	Modern Networking Practical	2
PSIT2P3	Microservices Architecture Practical	2
PSIT2P4	Image Processing Practical	2
Total Credits		24

Prerequisites : Fundamentals of Networking

- Network – collection of nodes and links that cooperate for communication
- Nodes – computer systems
 - Internal (routers, bridges, switches)
 - Terminal (workstations)
- Links – connections for transmitting data
- Protocol – standards for formatting and interpreting data and control information

What is a Network?

- A network refers to two or more connected computers that can share resources such as data, a printer, an Internet connection, applications, or a combination of these resources.

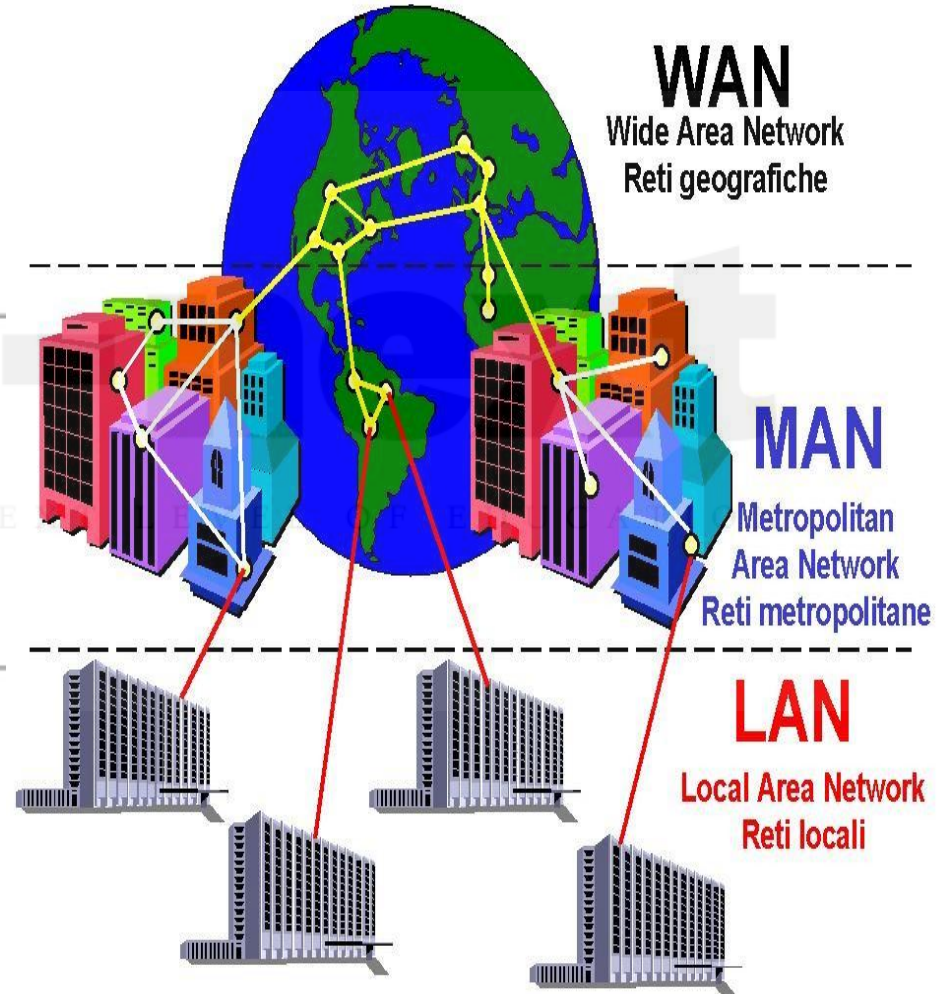


Types of Networks

Local Area Network (LAN)

Metropolitan Area Network (MAN)

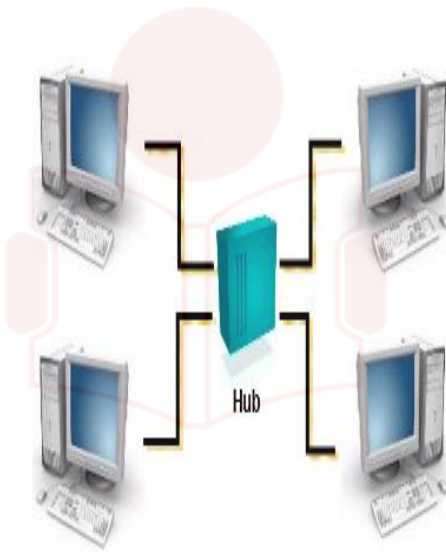
Wide Area Network (WAN)



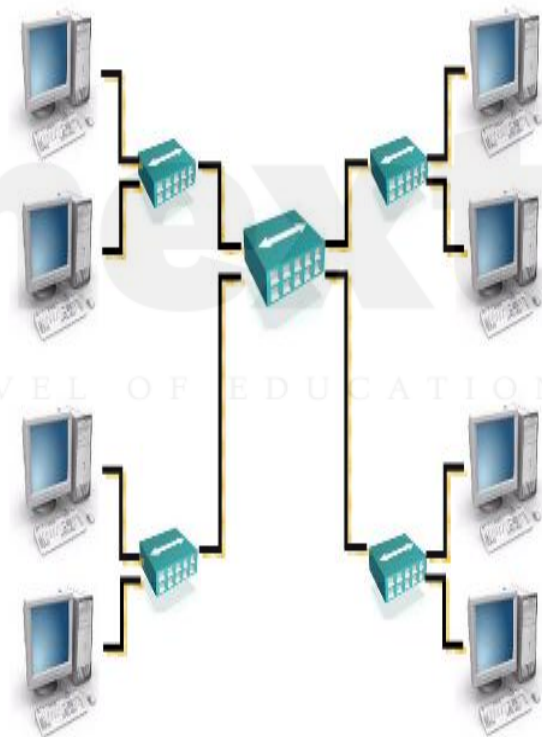
Network Topologies: Bus Topology



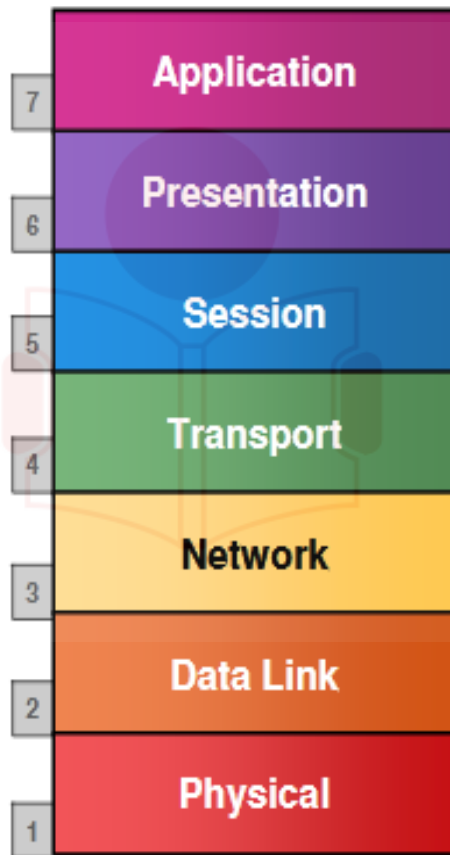
Network Topologies: Star Topology



Network Topologies: Extended Star Topology

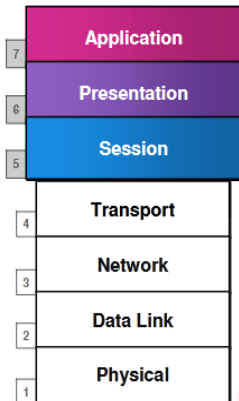


The OSI Model— Why a Layered Network Model?



- Reduces complexity
- Standardizes interfaces
- Facilitates modular engineering
- Ensures interoperable technology
- Accelerates evolution
- Simplifies teaching and learning

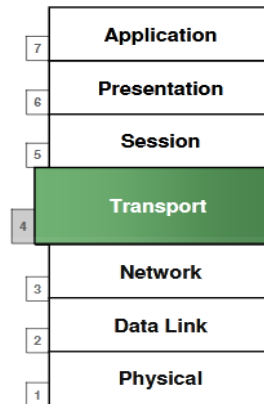
The Seven Layers of the OSI Model



Application Layers (Upper Layers):

- Network Processes to Applications
- Data Representation
- InterHost Communication

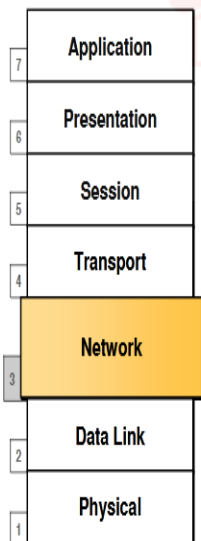
The Seven Layers of the OSI Model (Cont.)



End To End Connections:

- Handles transportation issues between hosts
- Ensures data transport reliability
- Establishes, maintains and terminates virtual circuits
- Provides reliability through fault detection and recovery
- Information flow control

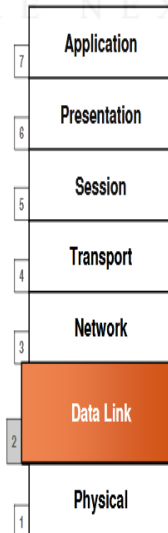
The Seven Layers of the OSI Model (Cont.)



Data Delivery:

- Provides connectivity and path selection between two host systems
- Routes data packets
- Selects best path to deliver data
- The Network layer prioritizes data known as Quality of Service (QoS)

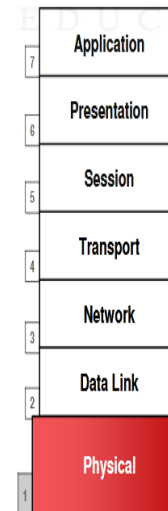
The Seven Layers of the OSI Model (Cont.)



Access to Media:

- Defines how data is formatted for transmission and how access to the network is controlled

The Seven Layers of the OSI Model (Cont.)

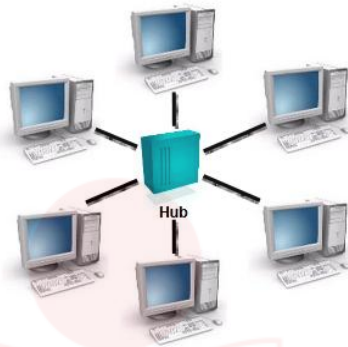


Binary Transmission:

- Defines the electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating the physical link

Hub or Repeater

- A hub (concentrator) is a device that repeats the signals it receives on one port to all other ports. It is a central connection point for several network devices.

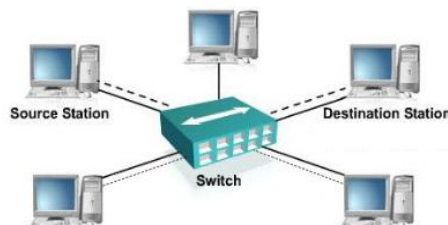


Network Interface Card



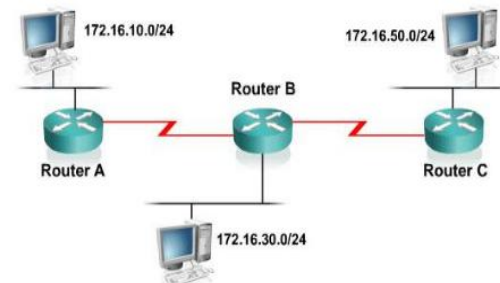
Switch

- When a switch receives data the switch examines the data link header for the MAC address of the destination station and forwards it to the correct port. This opens a path between ports that can use the full bandwidth of the topology.



Routers

- Routers facilitate communication within this internet work. It decides how to send packets within the network so that they arrive at their destination.

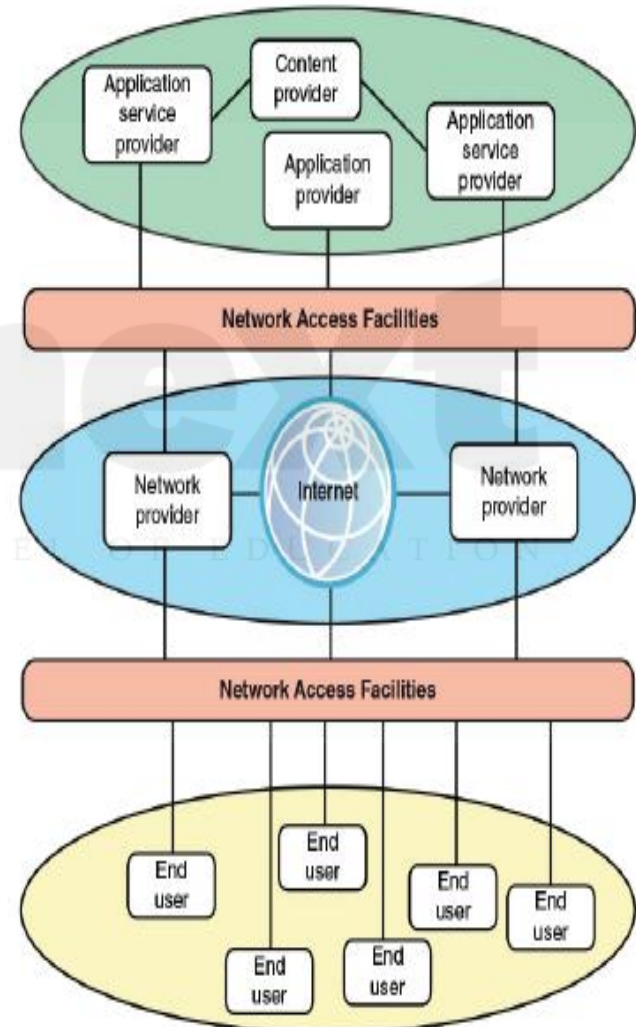




Elements of Modern Networking

The Networking Ecosystem

- Figure depicts the modern networking ecosystem in very general terms. The entire ecosystem exists to
- provide services to end users. The term **end user, or simply user, is used here as a very general term, to**
- encompass users working within an enterprise or in a public setting or at home. The user platform can be
- fixed (for example, PC or workstation), portable (for example, laptop), or mobile (for example, tablet or smartphone).



The Networking Ecosystem: End User

End User :

- The ultimate consumer of applications, data and services on a computing platform.



The Networking Ecosystem : Network Provider

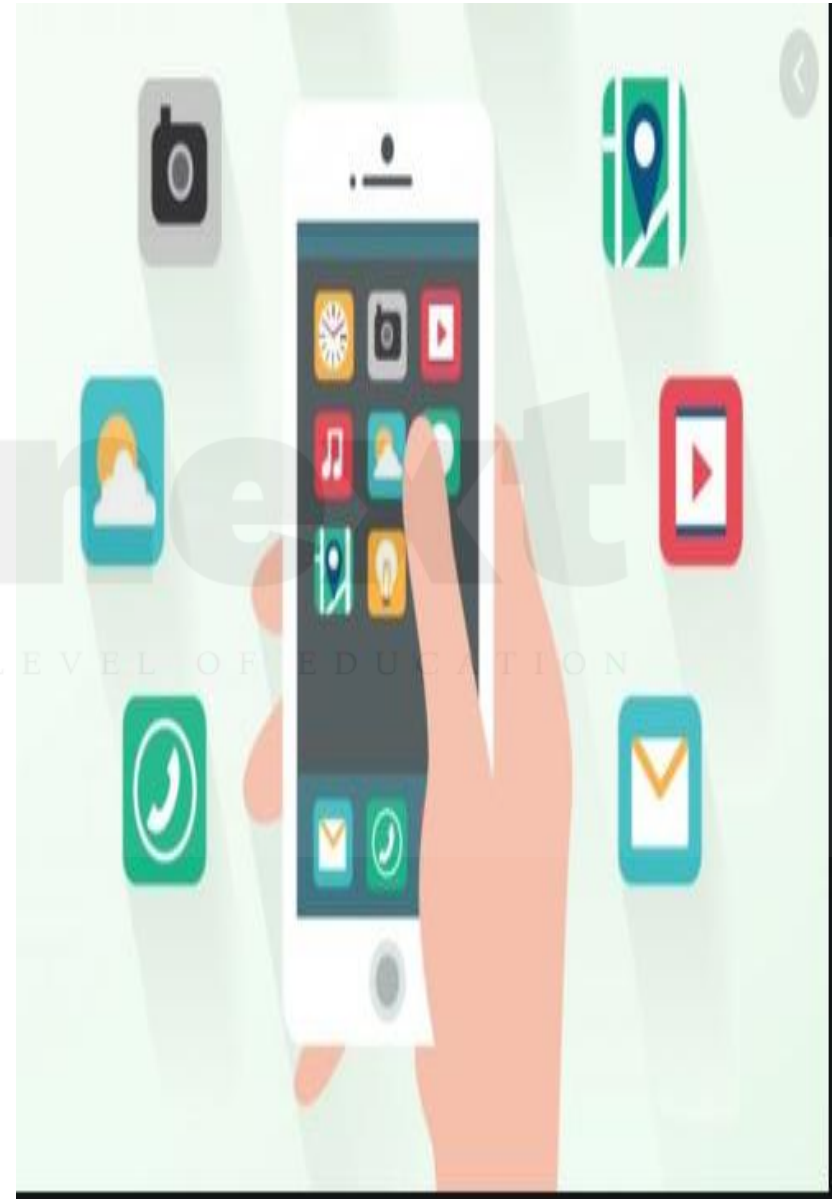
- An Organization that Delivers communications services over a typically large geographic area.
- It provides, maintains and manages network equipments and networks either public or private.
- Users connect to networks-based services and connect through a wide variety of network access facilities.
- These includes Digital Subscriber line(DSL) and Cable Modems, Wi-Fi and Microwave Access(WiMax) wireless Modems and cellular modems.



The Networking Ecosystem : Application

Provider

- **Application providers provide applications, or apps,** that run on the user's platform, which is typically a mobile platform.
- An entity generating/selling user applications to be executed on the user's platform.
- A distinct category of provider is the **application service provider.**
- **Whereas the application provider** downloads software to the user's platform, the application service provider acts as a server or host of
- application software that is executed on the provider's platforms.
- Traditional examples of such software include web servers, e-mail servers, and database servers.
- The most prominent example now is the cloud computing Provider.



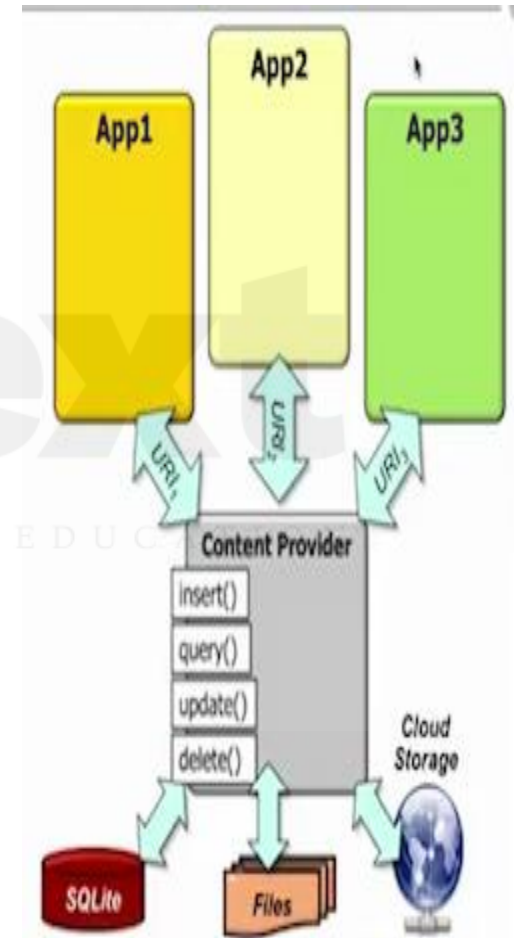
The Networking Ecosystem : Application Service Provider

- An organization that hosts software applications within its own facilities.
- It provides network-accessible applications such as e-mail, web hosting, banking, and cloud-based services.
- **A content provider serves the data to be consumed on the user device** (for example, e-mail, music, video).
- This data may be commercially provided intellectual property.
- In some instances, an enterprise may be an application or content provider.
- Examples of content providers are music record labels and movie studios.



The Networking Ecosystem : Content Provider

- An organization or individual that creates information, including educational or entertainment content, distributed via the Internet or enterprise networks.
- A content provider may or may not provide the software used to access the material.
- It is worth pointing out here two major elements of modern networking not explicitly depicted in this figure:
 - Data Center networking
 - IoT or Fog Networking



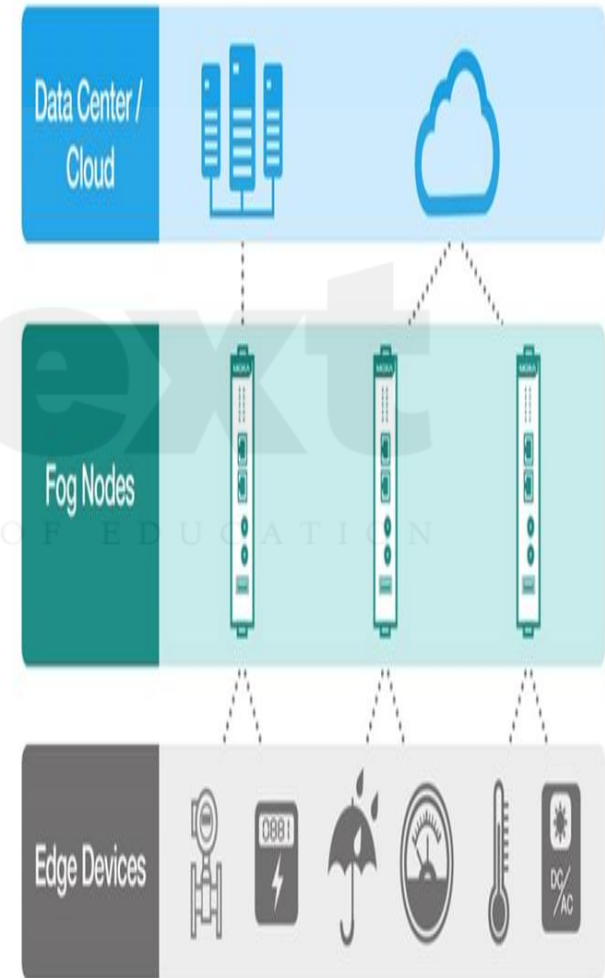
The Networking Ecosystem : Content Provider

Data center networking:

- Both large enterprise data centers and cloud provider data centers consist of very large numbers of interconnected servers.
- Typically, as much as 80 percent of the data traffic is within the data center network, and only 20 percent relies on external networks to reach users.

IoT or fog networking:

- An Internet of Things deployed by an enterprise may consist of hundreds, thousands, even millions of devices.
- The vast bulk of the data traffic to and from these devices is machine to machine, rather than user to machine.



Example Network Architectures

- This section introduces two example network architectures, and with them some of the networking terminology in common use :
 - A global Network Architecture
 - A typical Network hierarchy



Global Network Architecture

router

- A network device that forwards data packets from one network to another.
- The forwarding decision is based on network layer information and routing tables, often constructed by routing protocols.
- Routers require packets formatted in a routable protocol, the global standard being the Internet Protocol (IP).
- At the center of the figure is an IP backbone, or core, network, which could represent a portion of the Internet or an enterprise IP network.
- Typically, the backbone consists of high-performance routers, called **core routers**, **interconnected with high-volume optical links**.
- **The optical links often make use of what is known as wavelength-division multiplexing (WDM), such that each link has multiple logical channels occupying different portions of the optical bandwidth.**

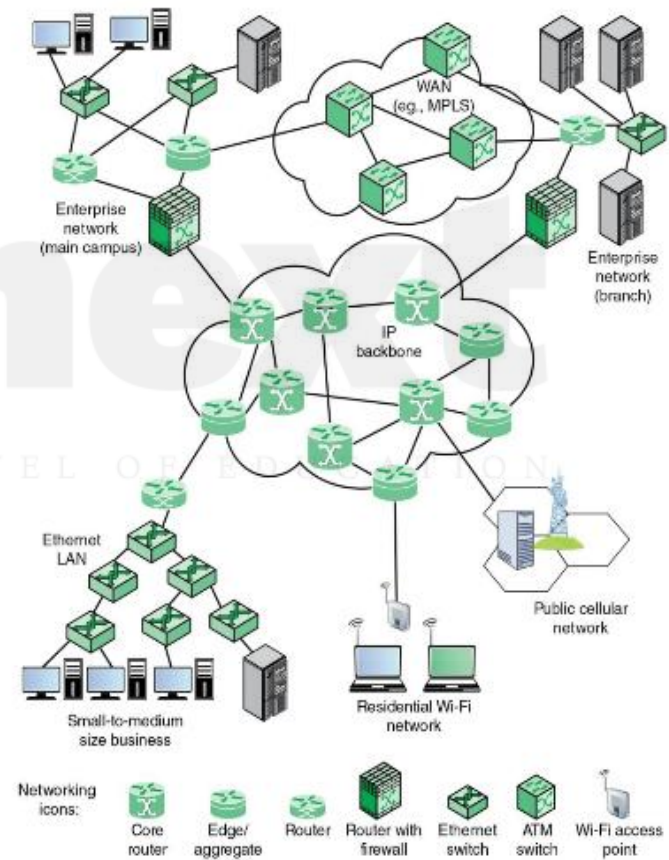


FIGURE 1.2 A Global Networking Architecture

Global Network Architecture

Core router :

- A router that resides within the middle of the network rather than at its periphery.
- The routers that make up the backbone of the Internet are core routers.
- At the periphery of an IP backbone are routers that provide connectivity to external networks and users.
- These routers are sometimes referred to as **edge routers or aggregation routers. Aggregation routers are also** used within an enterprise network to connect a number of routers and switches, to external resources, such as an IP backbone or a high-speed WAN.

Global Network Architecture

- **edge router :**
- A router that sits at the periphery of a network. Also called an access router or aggregation router.
- The upper part of Figure depicts a portion of what might be a large enterprise network.
- The figure shows two sections of the network connected via a private high-speed WAN, with switches interconnected with optical links.
- MPLS using IP is a common switching protocol used for such WANs; wide-area Ethernet is another option.
- Enterprise assets are connected to, and protected from, an IP backbone or the Internet via routers with firewall capability, a not uncommon arrangement for implementing the firewall.
- The lower left of the figure depicts what might be a layout for a small- or medium-size business, which relies on an Ethernet LAN.
- Connection to the Internet through a router could be through a cable or DSL connection or a dedicated high-speed link.
- The lower portion of Figure also shows an individual residential user connected to an Internet service provider (ISP) through some sort of subscriber connection.

Global Network Architecture

- Common examples of such a connection are a DSL, which provides a high-speed link over telephone lines and requires a special DSL modem, and a cable TV
- facility, which requires a cable modem, or some type of wireless connection.
- In each case, there are separate issues concerning signal encoding, error control, and the internal structure of the subscriber network.
- Finally, mobile devices, such as smartphones and tablets, can connect to the Internet through the public
- **Multiprotocol Label Switching (MPLS)**
- A protocol developed by IETF for directing packets in a wide-area IP network, or other WAN.
- MPLS adds a 32-bit label to each packet to improve network efficiency and to enable routers to direct packets along predefined routes in accordance with the required quality of service.

A typical Network hierarchy

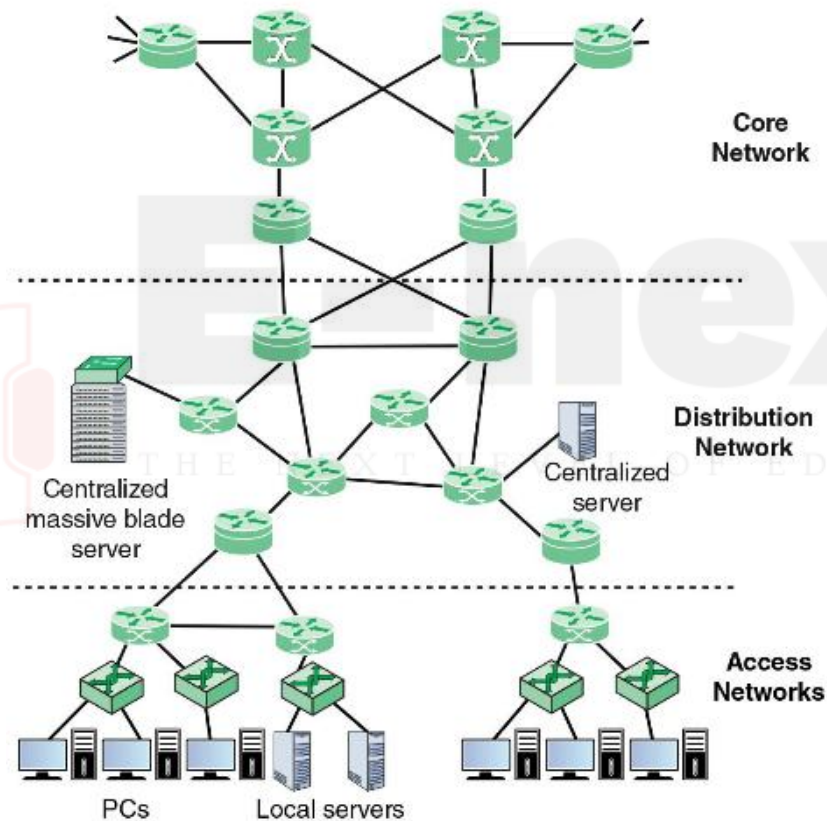


FIGURE 1.3 A Typical Network Hierarchy

A typical Network hierarchy

- **access network :**
- A network that connects directly to the end user or customer.
- **Layer 3 (L3) switch :**
- A high-performance device for network routing. Layer 3 switches are very similar to routers.
- The key difference between L3 switches and routers is that a L3 switch replaces some of a router's software logic with hardware to offer better performance.
- L3 switches often cost less than traditional routers.
- Designed for use
- within LANs, a L3 switch will typically not possess the WAN ports and WAN features a traditional router has.
- One or more access routers connect the local assets to the next higher level of the hierarchy, the distribution network.

A typical Network hierarchy

- Designed for use within LANs, a L3 switch will typically not possess the WAN ports and WAN features a traditional router has.
- One or more access routers connect the local assets to the next higher level of the hierarchy, the distribution network.
- This connection may be via the Internet or some other public or private communications facility.
- Thus, as described in the preceding subsection, these access routers function as edge routers that forward traffic into and out of the access network.

A typical Network hierarchy

- **distribution network:**
- The **distribution network** connects access networks with each other and with the core network.
- An edge router in the distribution network connects to an edge router in an access network to provide connectivity.
- The two routers are configured to recognize each other and will generally exchange routing and connectivity information and, typically, some traffic-related information.
- The **distribution network** also have serves to aggregate traffic destined for the core router, which protects the core from high-density peering.
- The use of a distribution network limits the number of routers that establish peer relationships with edge routers in the core, saving memory, processing, and transmission capacity.
- A distribution network may also directly connect servers that are of use to multiple **access networks**, such as database servers and network management servers.

A typical Network hierarchy

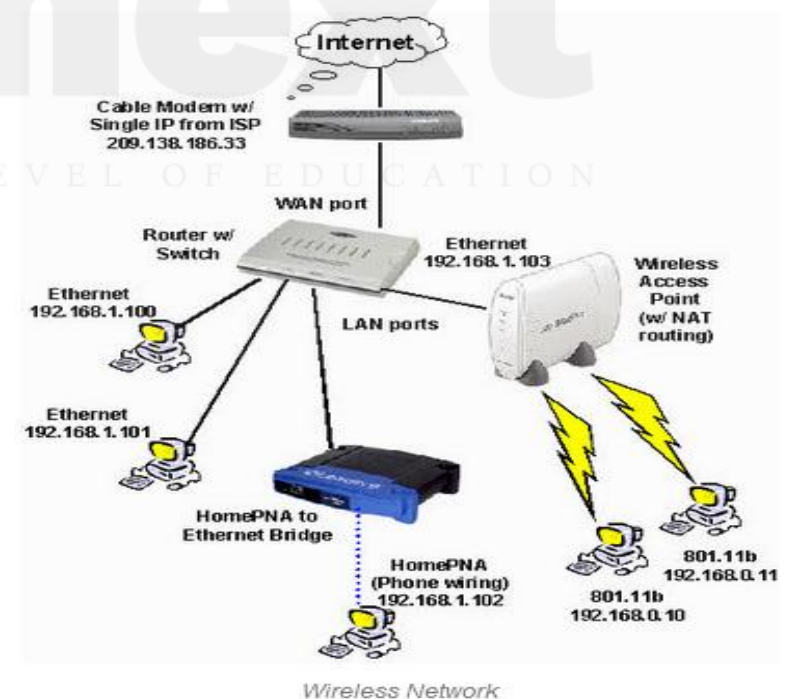
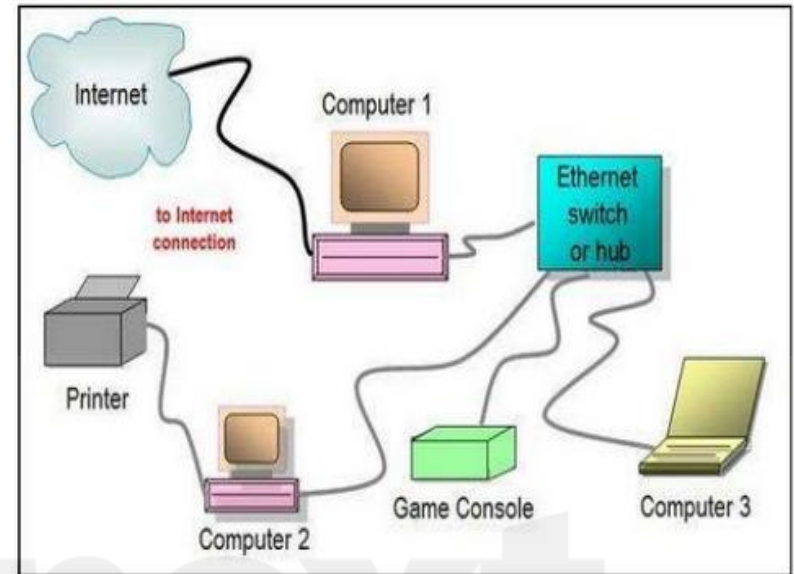
- **peering**
- An agreement between two routers to accept each other's data packets and forward them.
- A peer relationship generally involves the exchange of routing information.
- Again, as with access networks, some of the distribution routers may be purely internal and do not provide an edge router function.

A typical Network hierarchy

- **core/backbone network**
- A **central network** that provides networking services to attached distribution and access networks. Also referred to as a backbone network.
- The **core network**, also referred to as a **backbone network**, connects geographically dispersed **distribution networks** as well as providing access to other networks that are not part of the enterprise network.
- Typically, the core network will use very high performance routers, high-capacity transmission lines, and multiple interconnected routers for increased redundancy and capacity.
- The **core network** may also connect to high performance, high-capacity servers, such as large database servers and private cloud facilities.

Ethernet

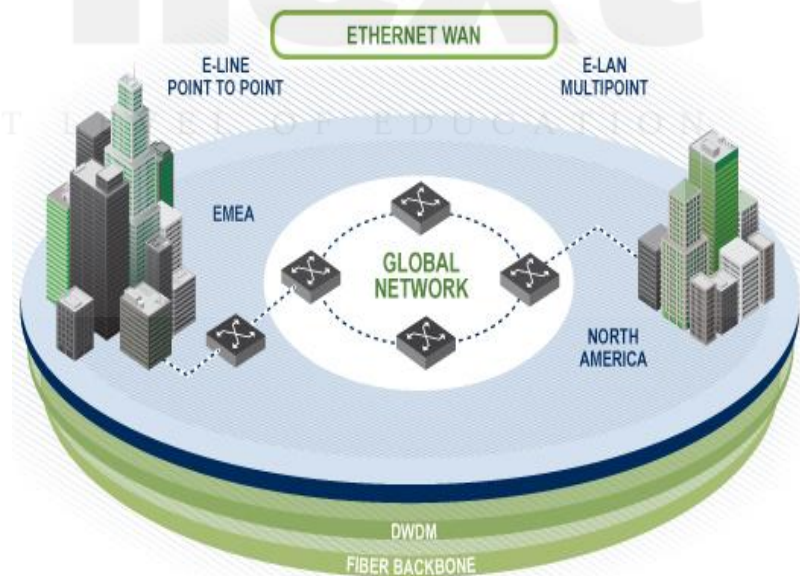
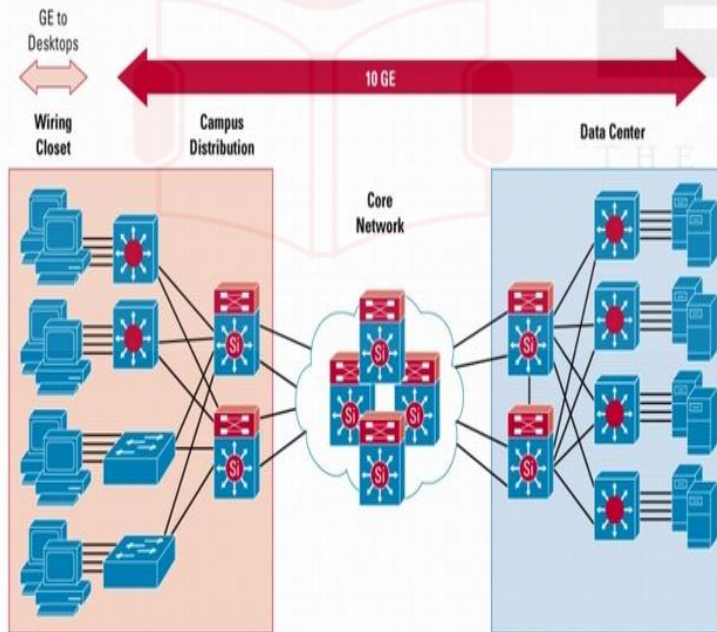
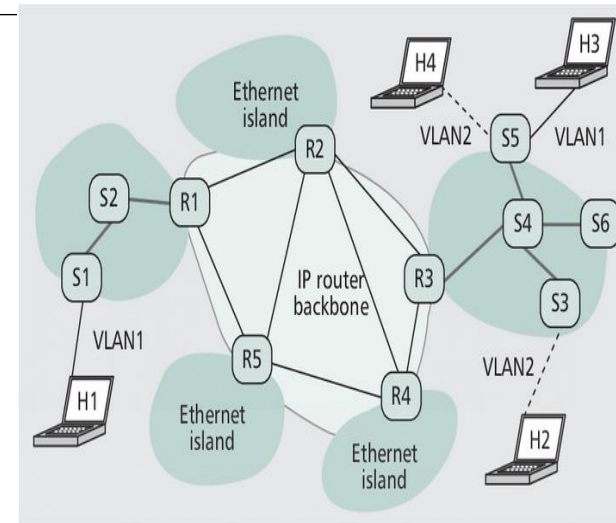
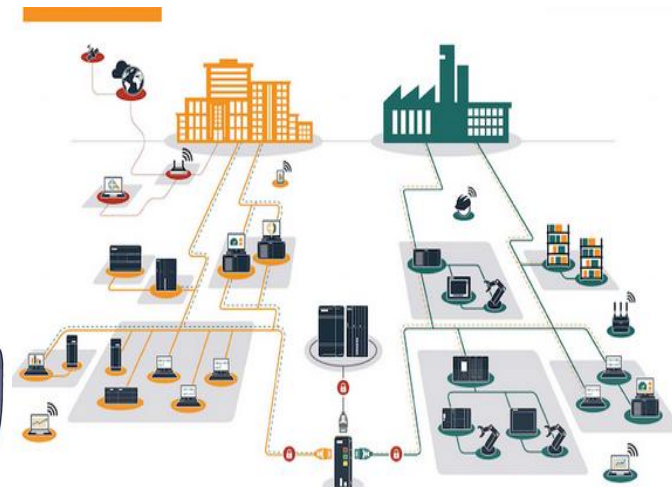
- Ethernet is most widely used LAN Technology, which is defined under IEEE standards 802.3.
- The reason behind its wide usability is Ethernet is easy to understand, implement, maintain and allows low-cost network implementation.
- Also, Ethernet offers flexibility in terms of topologies which are allowed. Ethernet generally uses Bus Topology.
- Ethernet operates in two layers of the OSI model, Physical Layer, and Data Link Layer.



Ethernet : Applications of Ethernet

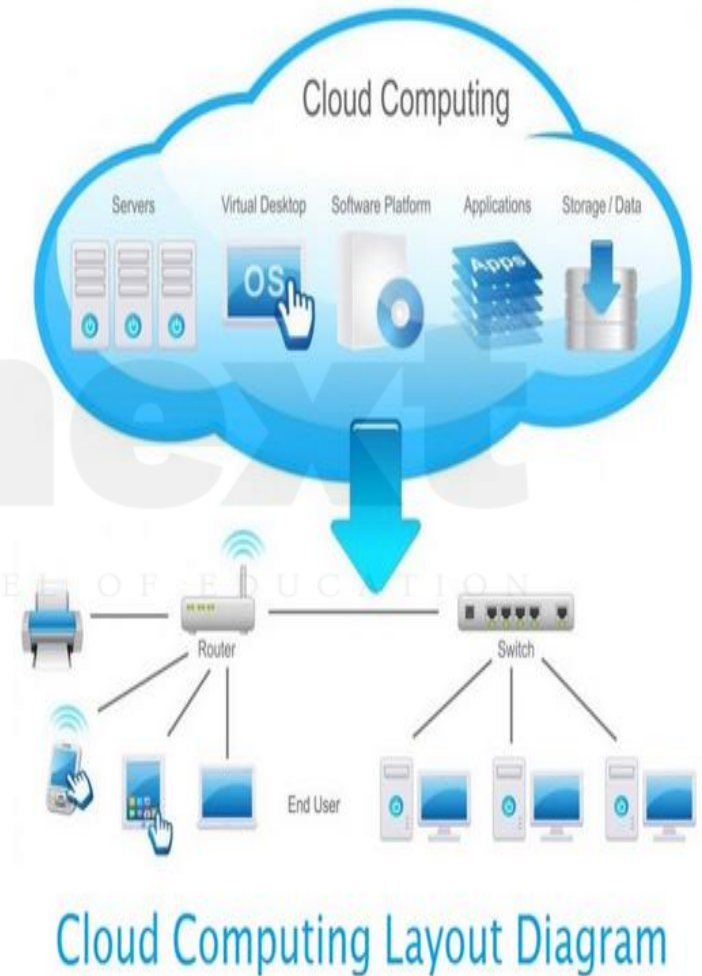
- **Ethernet in the Home**
- **Ethernet in the Office**
- **Ethernet in the enterprise**
- **Ethernet in the Data Center**
- **Ethernet for Wide-Area Networking**





Cloud Computing concepts

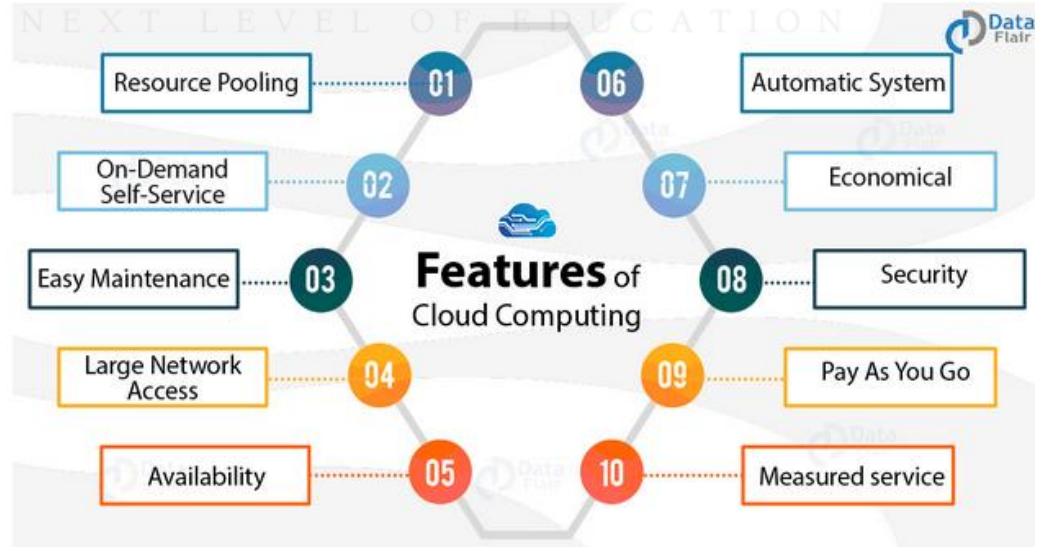
- Although the general concepts for cloud computing go back to the 1950s, cloud computing services first became available in the early 2000s, particularly targeted at large enterprises.
- Since then, cloud computing has spread to small- and medium-size businesses, and most recently to consumers.
- Apple's iCloud was launched in 2012 and had 20 million users within a week of launch.
- Evernote, the cloud-based note-taking and archiving service, launched in 2008, approached 100 million users in less than six years.
- In late 2014, Google announced that Google Drive had almost a quarter of a billion active users.
- Here, we look at the key elements of clouds, including cloud computing, cloud networking, and cloud storage.



Cloud Computing concepts :

Characteristics of cloud computing

- **Broad network access:**
- Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (for example, mobile phones, laptops, and personal digital assistants [PDAs]) and other traditional or cloud-based software services.



Cloud Computing concepts :

Characteristics of cloud computing

- **Rapid elasticity:**
- Cloud computing enables you to expand and reduce resources according to your specific service requirement.
- For example, you may need a large number of server resources for the duration of a specific task. You can then release these resources upon completion of the task.
- **Measured service:**
- Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (for example, storage, processing, bandwidth, and active user accounts).
- Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

Cloud Computing concepts :

Characteristics of cloud computing

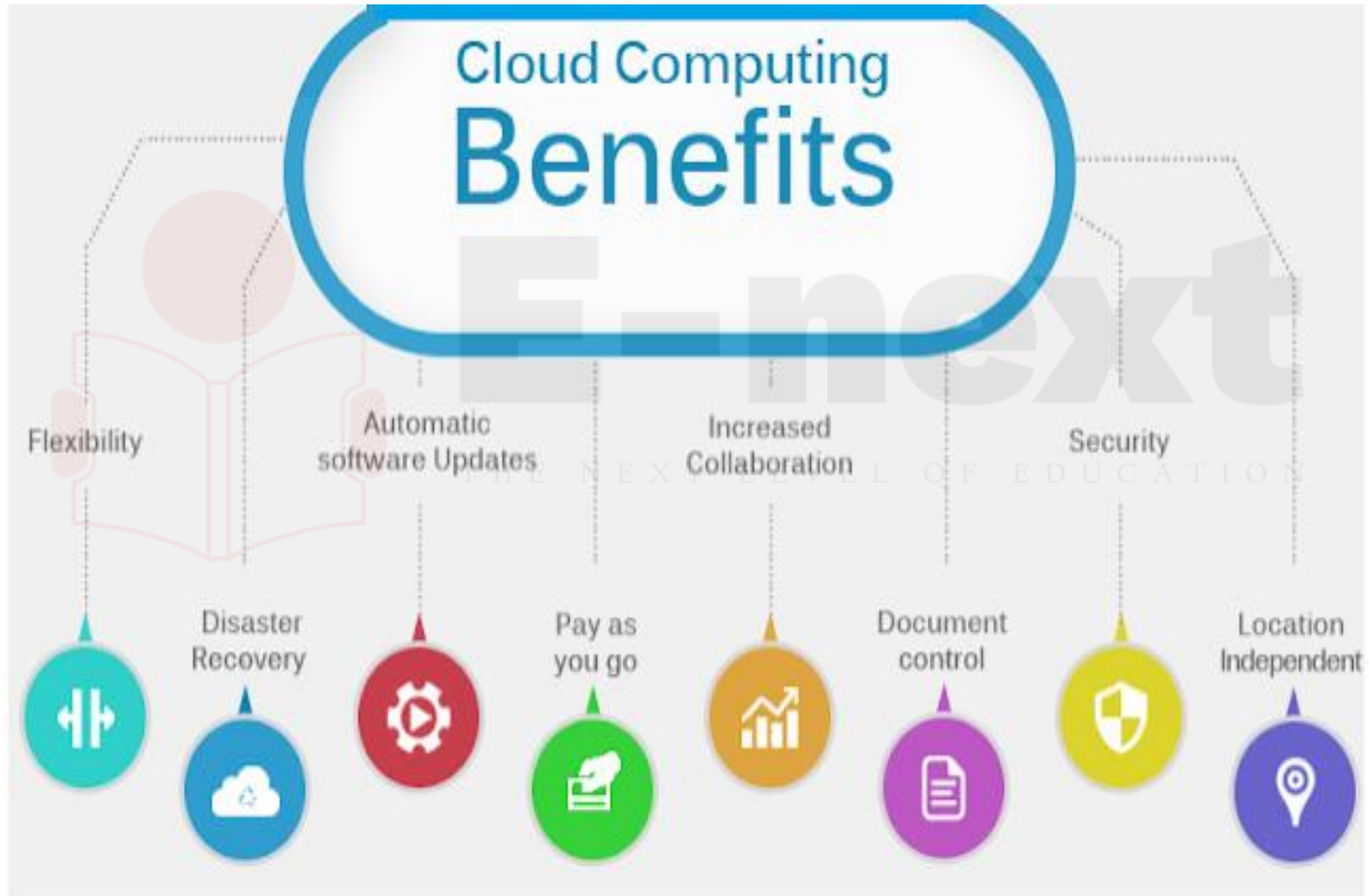
- **On-demand self-service:**
- A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- Because the service is on demand, the resources are not permanent parts of your IT infrastructure.

Cloud Computing concepts :

Characteristics of cloud computing

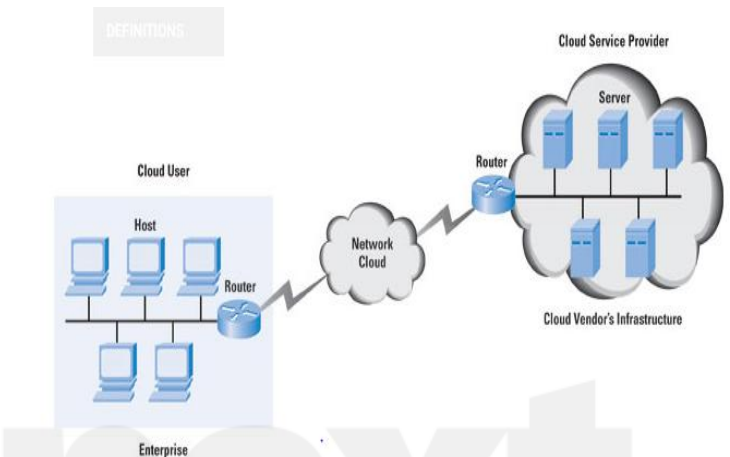
- **Resource pooling:**
- The provider's computing resources are pooled to serve multiple consumers using a multitenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- There is a degree of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources, but may be able to specify location at a higher level of abstraction (for example, country, state, or data center).
- Examples of resources include storage, processing, memory, network bandwidth, and virtual machines.
- Even private clouds tend to pool resources between different parts of the same organization.

The Benefits of Cloud Computing



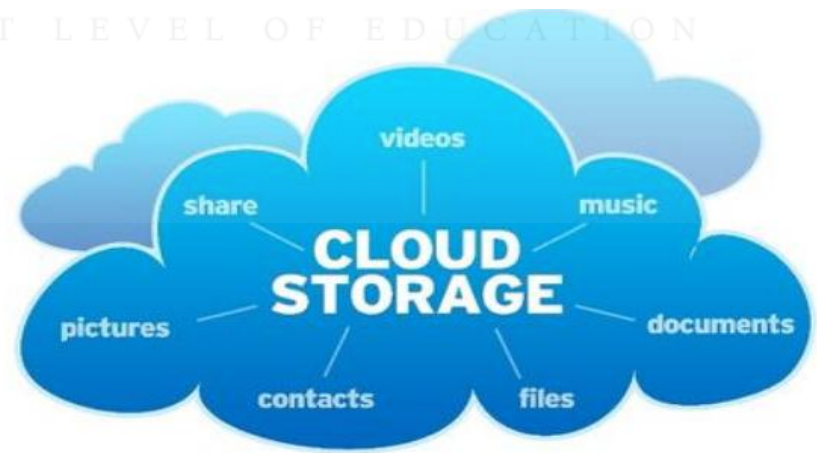
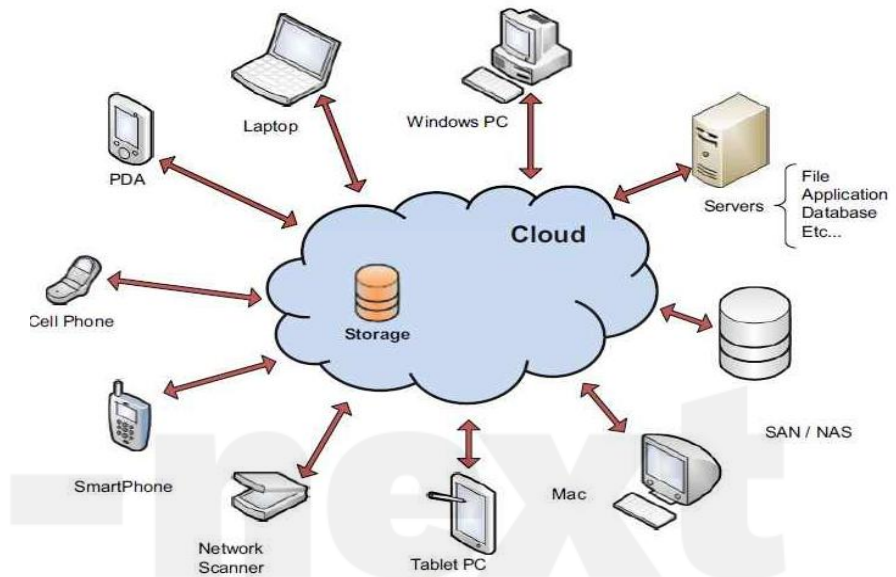
Cloud Networking

- Cloud networking refers to the networks and network management functionality that must be in place to enable cloud computing.
- Many cloud computing solutions rely on the Internet, but that is only a piece of the networking infrastructure.
- One example of cloud networking is the provisioning high-performance/high-reliability networking between the provider and subscriber.
- In this case, some or all of the traffic between an enterprise and the cloud bypasses the Internet and uses dedicated private network facilities owned or leased by the cloud service provider.
- More generally, cloud networking refers to the collection of network capabilities required to access a cloud, including making use of specialized services over the Internet, linking enterprise data centers to a cloud, and using firewalls and other network security devices at critical points to enforce access security



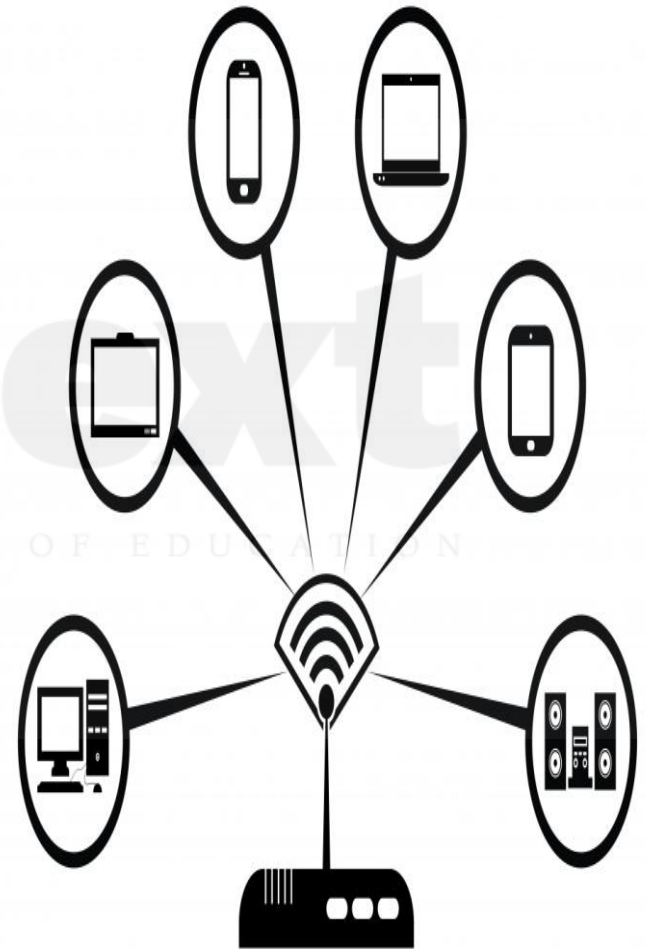
Cloud Storage

- We can think of cloud storage as a subset of cloud computing. In essence, cloud storage consists of database storage and database applications hosted remotely on cloud servers.
- Cloud storage enables small businesses and individual users to take advantage of data storage that scales with their needs and to take advantage of a variety of database applications without having to buy, maintain, and manage the storage assets.



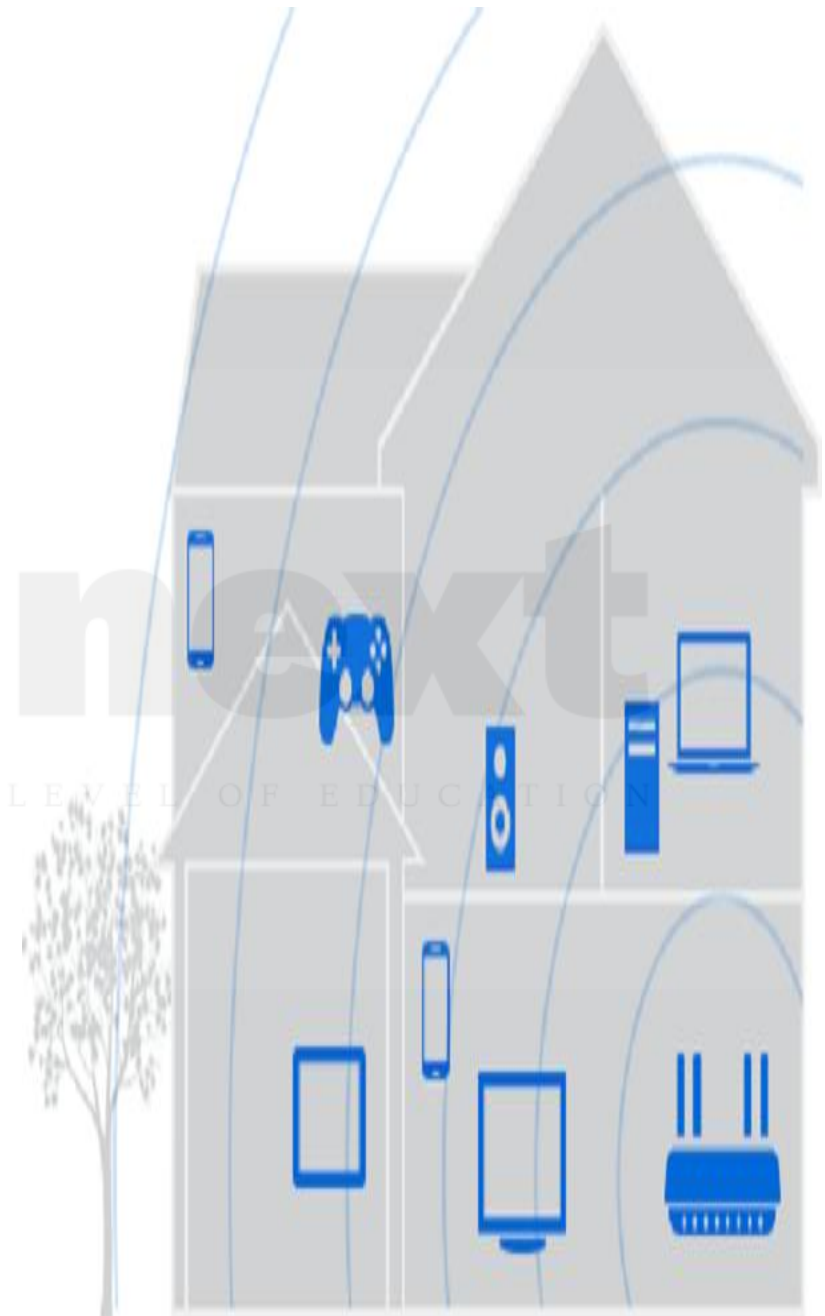
Wi-Fi

- Just as Ethernet has become the dominant technology for wired LANs, so WiFi, standardized by the IEEE 802.11 committee, has become the dominant technology for wireless LANs.
- Refers to the wireless LAN technology standardized by the IEEE 802.11 committee.
- The term WiFi designates products that have been certified by the Wi-Fi Alliance to conform to the 802.11 standards and have passed interoperability tests.
- **Applications of Wi-Fi**
 - Wi-Fi is the predominant wireless Internet access technology, used in homes, offices, and public spaces.
 - Wi-Fi in the home now connects computers, tablets, smartphones, and a host of electronic devices, such as video cameras, TVs, and thermostats.
 - Wi-Fi in the enterprise has become an essential means of enhancing worker productivity and network effectiveness.



Wi-Fi in the home

- The first important use of Wi-Fi in the home was to replace Ethernet cabling for connecting desktop and laptop computers with each other and with the Internet.
- A typical layout is a desktop computer with an attached router/modem that provides an interface to the Internet.
- Wi-Fi greatly simplified the hookup.
- Not only is there no need for a physical cable hookup, but the laptops can be moved easily from room to room or even outside the house.
- Because both Wi-Fi and cellular capability are now standard on both smartphones and tablets, the home Wi-Fi provides a costeffective way to the Internet.
- The smartphone or tablet will automatically use a Wi-Fi connection to the Internet if available, and only switch to the more expensive cellular connection if the Wi-Fi



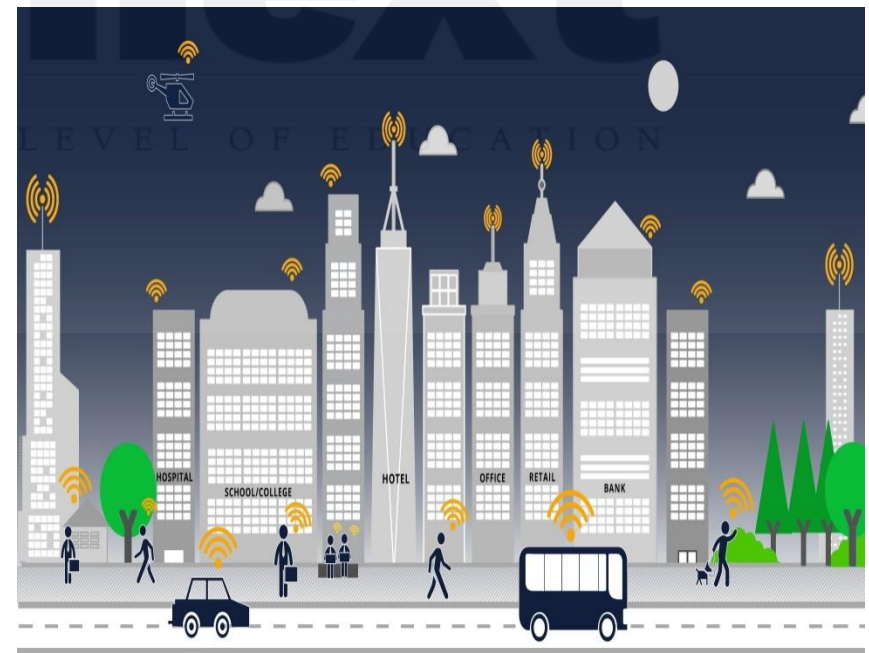
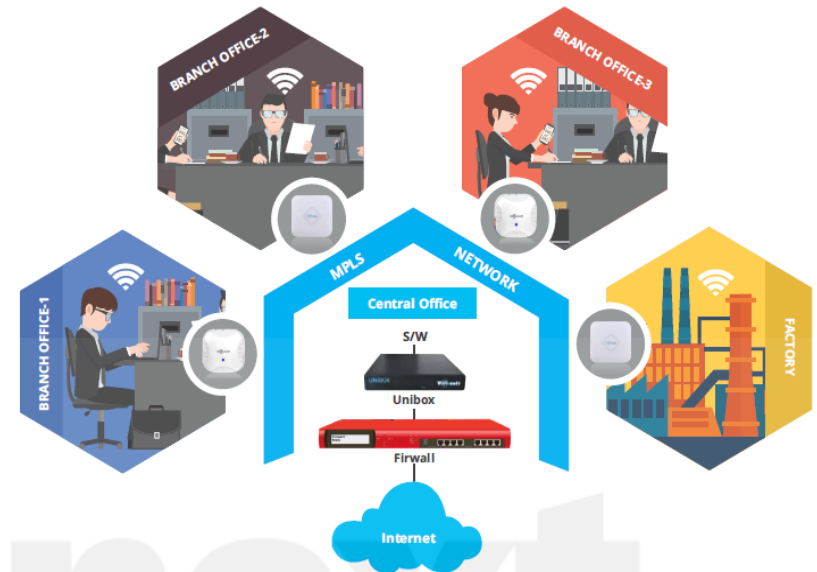
Public Wi-Fi

- Access to the Internet via Wi-Fi has expanded dramatically in recent years, as more and more facilities provide
- a Wi-Fi hotspot, which enables any Wi-Fi device to attach.
- Wi-Fi hotspots are provided in coffee shops, restaurants, train stations, airports, libraries, hotels, hospitals, department stores, RV parks, and many other places.
- So many hotspots are available that it is rare to be too far from one.
- There are now numerous tablet and smartphone apps that increase their convenience.
- Even very remote places will be able to support hotspots with the development of the satellite Wi-Fi hotspot.
- The first company to develop such a product is the satellite communications company Iridium.



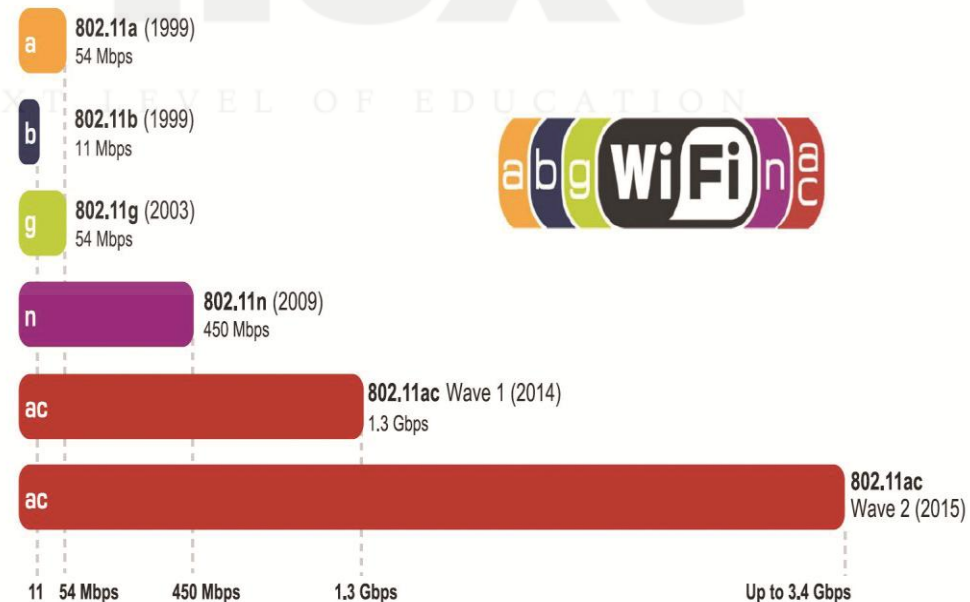
Enterprise Wi-Fi

- Wi-Fi connections to the enterprise network have been offered by many organizations of all sizes, including public and private sector.
- **Two trends have driven the transition to a Wi-Fi centered enterprise.**
- **First**, the demand has increased, with more and more employees preferring to use laptops, tablets, and smartphones to connect to the enterprise network, rather than a desktop computer.
- **Second**, the arrival of Gigabit Ethernet, especially the IEEE 802.11ac standard, allows the enterprise network to support high-speed connections to many mobile devices simultaneously.
- Enterprises accepted the need for, and then began to encourage, the practice known as bring your own device (BYOD).
- Employees can use the same devices and the same applications to continue their work or check their e-mail from wherever they are—home, at their local coffee shop, or while traveling



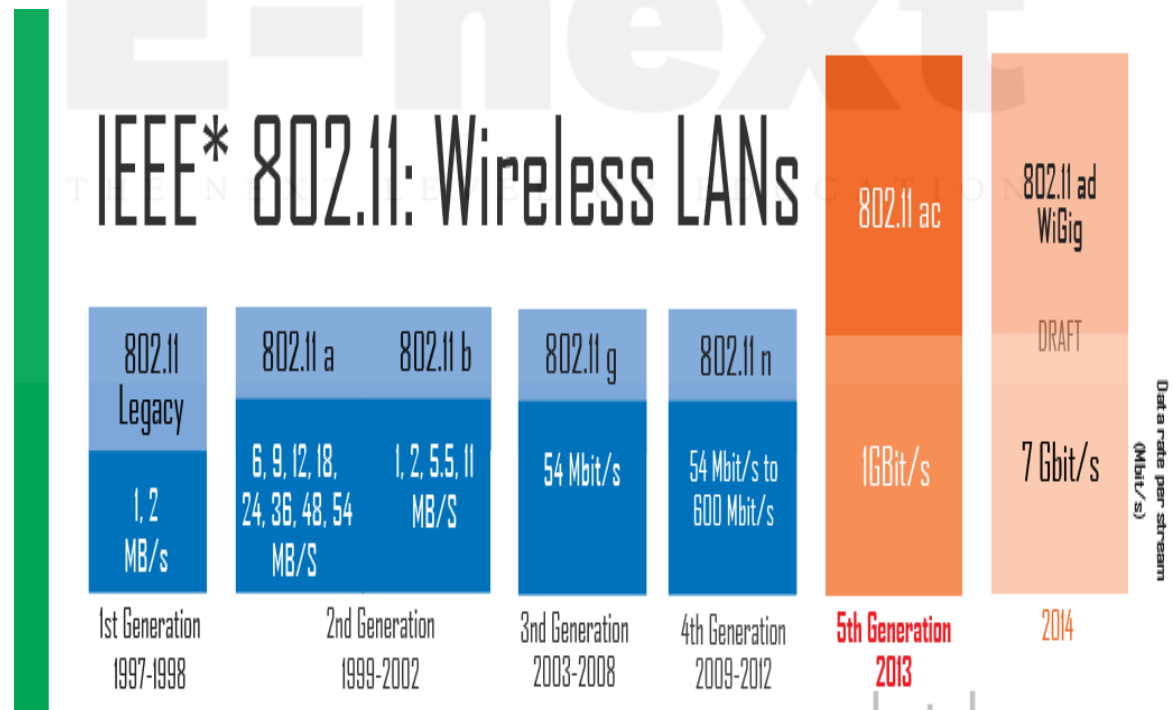
Wi-Fi Standards

- Essential to the success of Wi-Fi is interoperability.
- Wi-Fi-enabled devices must be able to communicate with Wi-Fi access points, such as the home router, the enterprise access point, and public hotspots, regardless of the manufacturer of the device or access point.
- **Such interoperability is guaranteed by two organizations.**
 - **First**, the IEEE 802.11 wireless LAN committee develops the protocol and signaling standards for Wi-Fi.
 - **Second** the Wi-Fi Alliance creates test suites to certify interoperability for commercial products that conform to various IEEE 802.11 standards.
- The term WiFi (wireless fidelity) is used for products certified by the Alliance.
- IEEE 802.11 Wireless LAN Working Group



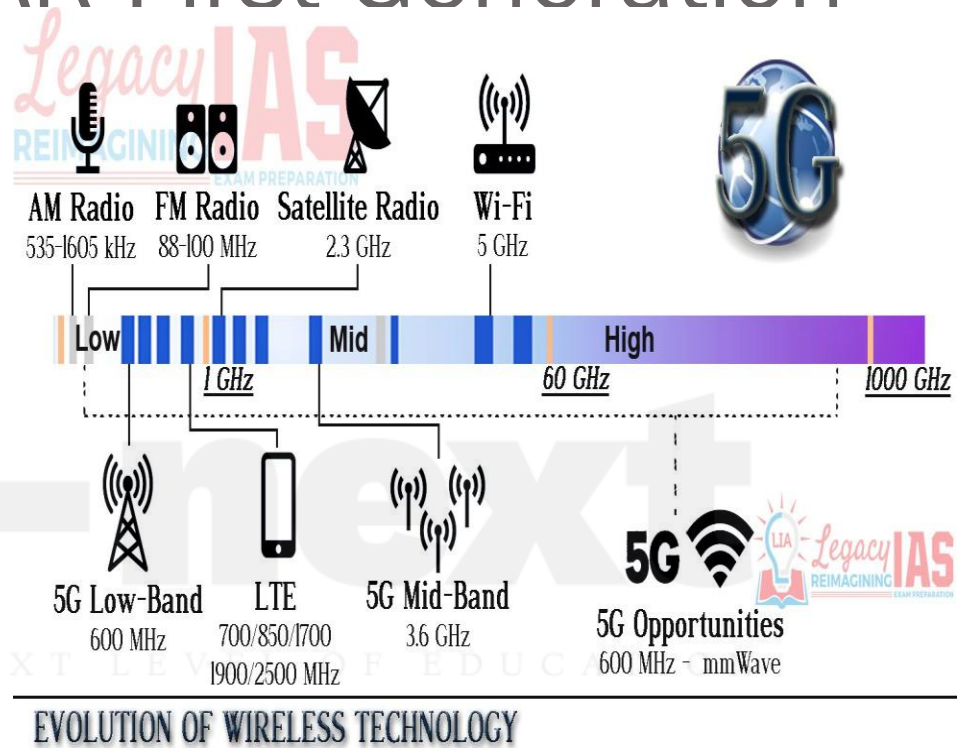
Wi-Fi Data Rates

- Just as businesses and home users have generated a need to extend the Ethernet standard to speeds in the gigabits per second (Gbps) range, the same requirement exists for Wi-Fi.
- As the technology of antennas, wireless transmission techniques, and wireless protocol design has evolved, the IEEE 802.11 committee has been able to introduce standards for new versions of Wi-Fi at ever-higher speeds.
- Once the standard is issued, industry quickly develops the products.
- Here's a brief chronology, starting with the original standard, which was simply called IEEE 802.11, and showing the maximum data rate for each version :
- 802.11 (1997): 2 Mbps (megabits per second, million bits per second)
- 802.11a (1999): 54 Mbps
- 802.11b (1999): 11 Mbps
- 802.11n (1999): 600 Mbps
- 802.11g (2003): 54 Mbps
- 802.11ad (2012): 6.76 Gbps

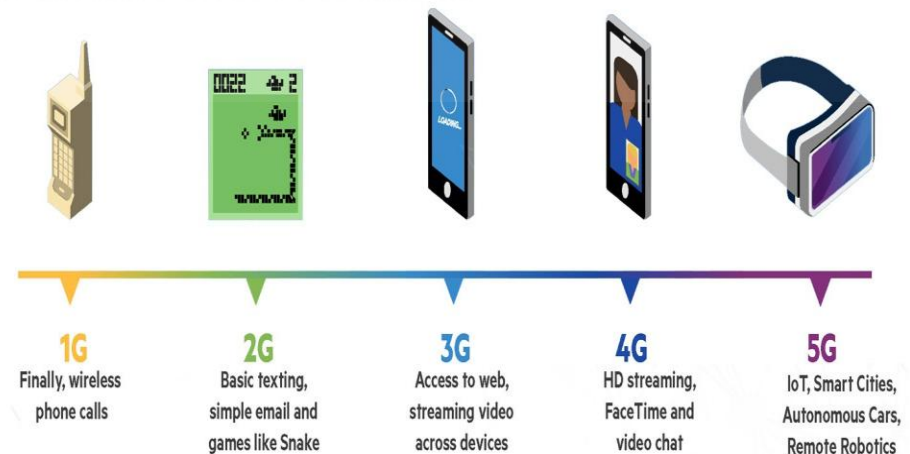


4G/5G CELLULAR First Generation

- Cellular technology is the foundation of mobile wireless communications and supports users in locations that are not easily served by wired networks.
- Cellular technology is the underlying technology for mobile telephones, personal communications systems, wireless Internet and wireless web applications, and much more.
- This section looks at how cellular technology has evolved through four generations and is poised for a fifth generation.



EVOLUTION OF WIRELESS TECHNOLOGY



First Generation

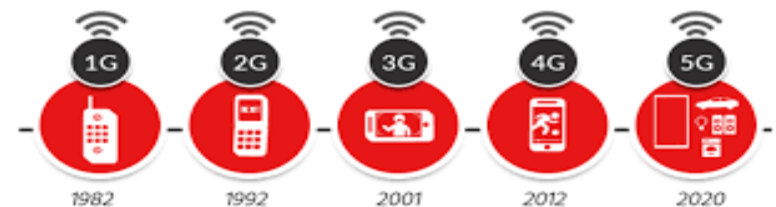
- The original cellular networks, now dubbed 1G, provided analog traffic channels and were designed to be an extension of the public switched telephone networks.
- Users with brick-sized cell phones placed and received calls in the same fashion as landline subscribers.
- The most widely deployed 1G system was the Advanced Mobile Phone Service (AMPS), developed by AT&T.
- Voice transmission was purely analog and control signals were sent over a 10-kbps analog channel.
- 1G systems send user traffic in the clear, providing no security.
- In 1G systems, each cell supports a number of channels. At any given time a channel is allocated to only one user.



Timeline of mobile communication generations
1G -5G

Second Generation

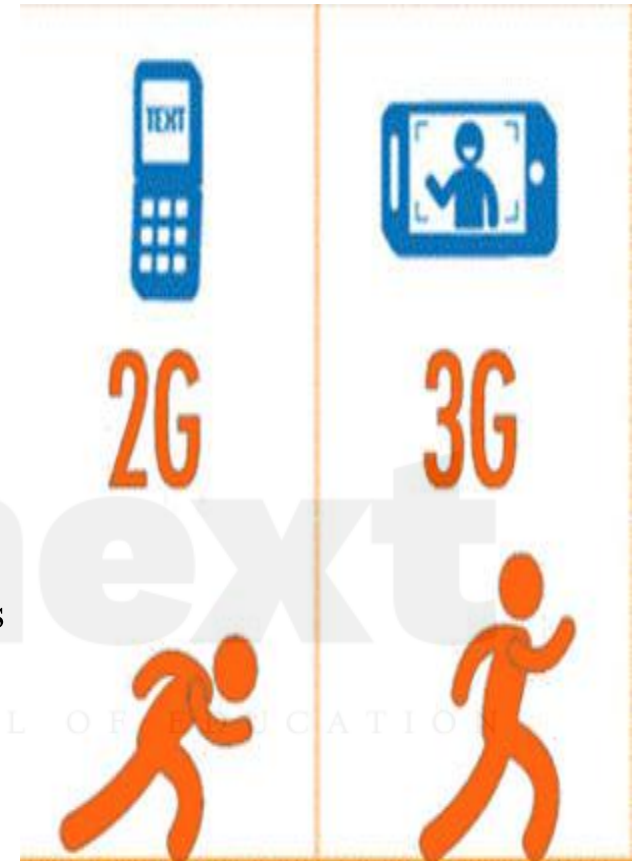
- First-generation cellular networks quickly became highly popular, threatening to swamp available capacity.
- Second-generation (2G) systems were developed to provide higher-quality signals, higher data rates for support of digital services, and greater capacity.
- **Key differences between 1G and 2G networks include the following:**
 - **Digital traffic channels:** 1G systems are designed to support voice channels; digital traffic is supported only by the use of a modem that converts the digital data into analog form. 2G systems provide digital traffic channels.
 - **Encryption:** Because all the user traffic, and the control traffic, is digitized in 2G systems, it is a relatively simple matter to encrypt all the traffic to prevent eavesdropping. All 2G systems provide this capability,
 - **Error detection and correction:** The digital traffic stream of 2G systems also lends itself to the use of error detection and correction techniques.
 - **Channel access:** as 1G the 2G systems also provide multiple channels per cell, but On the contrary 1G each



Timeline of mobile communication generations
1G -5G

Third Generation

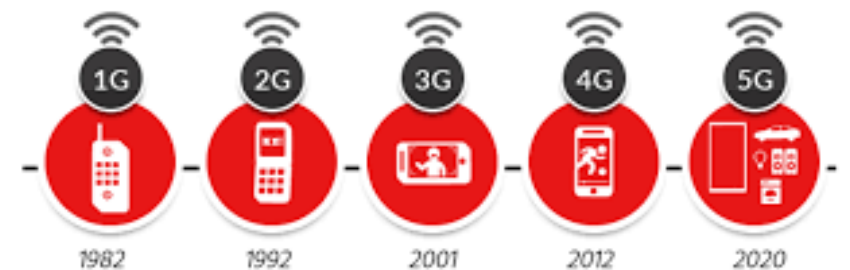
- The objective of the third generation (3G) of wireless communication is to provide fairly high-speed wireless communications to support multimedia, data, and video in addition to voice.
- **3G systems has the following features:**
 - **Bandwidth:** 3G systems has channel usage to 5 MHz .
 - **Data rate:** Target data rates are 144 and 384 kbps.
 - **Multirate:** The term multirate refers to the provision of multiple fixed-data-rate logical channels to a given user, in which different data rates are provided on different logical channels.
 - Further, the traffic on each logical channel can be switched independently through the wireless and fixed networks to different destinations.
 - The advantage of multirate is that the system can flexibly



Timeline of mobile communication generations
1G -5G

Fourth Generation

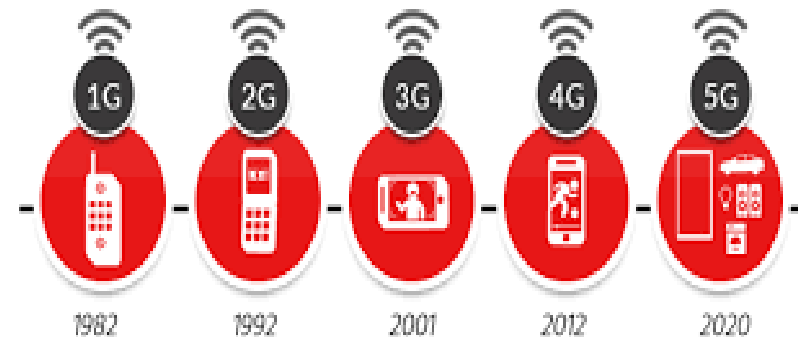
- The evolution of smartphones and cellular networks has ushered in a new generation of capabilities and standards, which is collectively called 4G.
- 4G systems provide ultra-broadband Internet access for a variety of mobile devices including laptops, smartphones, and tablets.
- 4G networks support Mobile web access and high-bandwidth applications such as highdefinition mobile TV, mobile video conferencing, and gaming services.
- **4G systems have the following characteristics:**
 - Based on an all-IP packet switched network
 - Support peak data rates of up to approximately 100 Mbps for high-mobility mobile access and up to approximately 1 Gbps for low-mobility access such as local wireless access
 - Dynamically share and use the network resources to support more simultaneous users per cell



Timeline of mobile communication generations
1G -5G

Fifth Generation

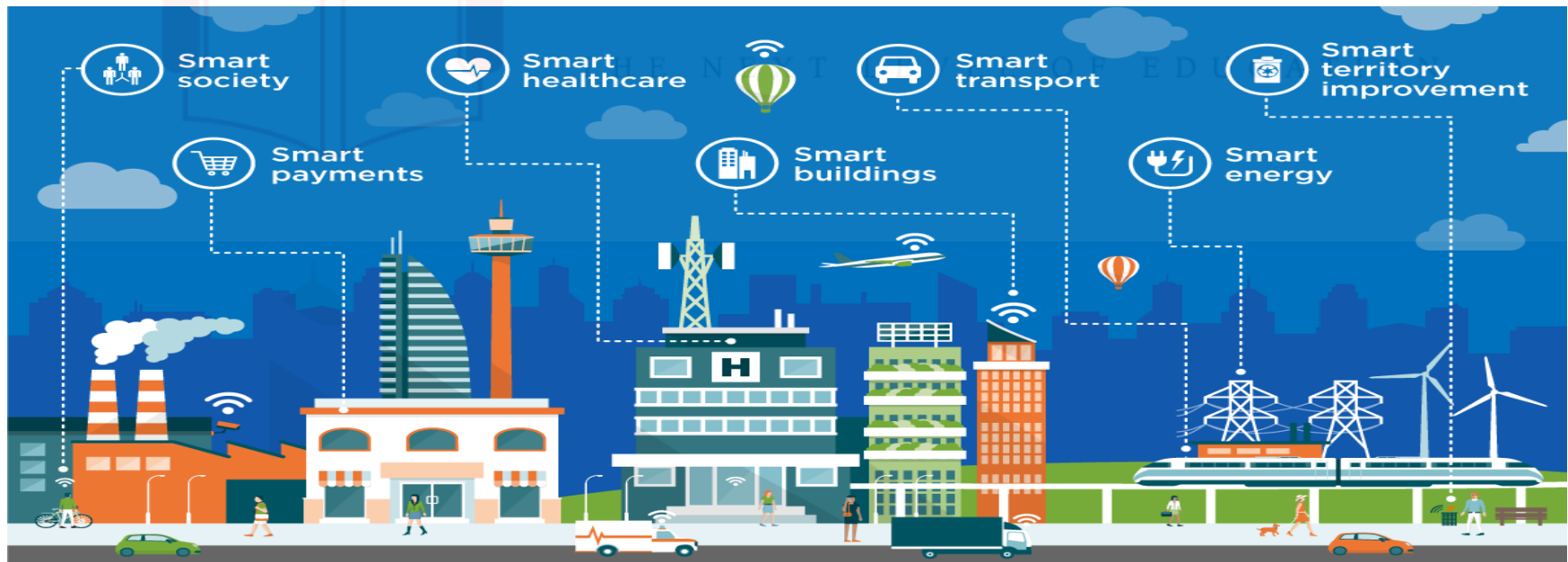
- 5G systems are still some years away (perhaps 2020), but 5G technologies are likely an area of active research.
- By 2020, the huge amounts of data traffic generated by tablets and smartphones will be augmented by an equally huge, and perhaps much larger, amount of traffic from the Internet of Things, which includes shoes, watches, appliances, cars, thermostats, door locks, and much more.
- With 4G, we may have reached a point of diminishing returns on network efficiency.
- There will be incremental improvements in the future, but significant increases in transmission efficiency seem unlikely.
- Instead, the focus for 5G will be on building more intelligence into the network, to meet service quality demands by dynamic use of priorities, adaptive network reconfiguration, and other network management techniques.



Timeline of mobile communication generations

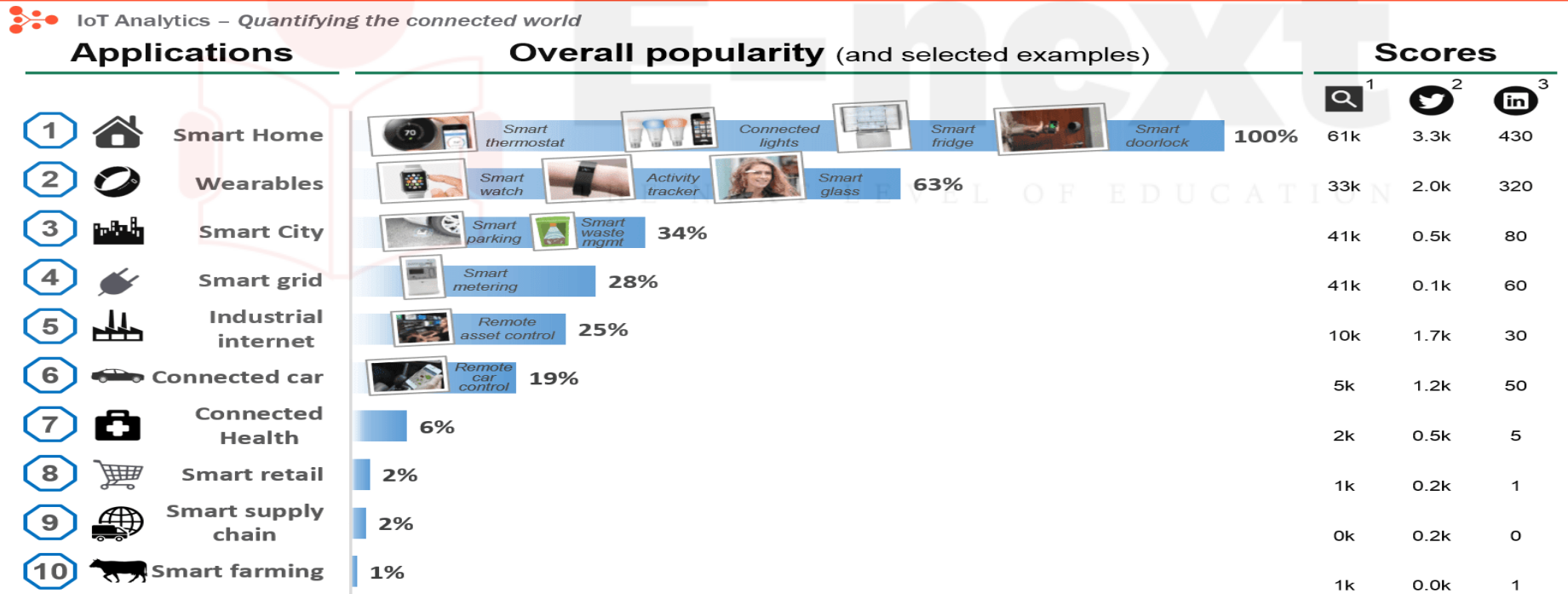
Internet Of Things(IOT)

- The Internet of Things (IoT) is the latest development in the long and continuing revolution of computing and communications.
- Its size, ubiquity, and influence on everyday lives, business, and government dwarf any technical advance that has gone before.
- This section provides a brief overview of the IoT, which is dealt with in greater detail later in the Slides.
- So Internet of Things (IoT) is the expanding connectivity, particularly via the Internet of a wide range of sensors, actuators, and other embedded systems.



Things on the Internet of Things

- The Internet of Things (IoT) is a term that refers to the expanding interconnection of smart devices, ranging from appliances to tiny sensors.
- A dominant theme is the embedding of short-range mobile transceivers into a wide array of gadgets and everyday items, enabling new forms of communication between people and things, and between things themselves.
- The Internet now supports the interconnection of billions of industrial and personal objects, usually through cloud systems.
- The objects deliver sensor information, act on their environment, and in some cases modify themselves, to create overall management of a larger system, like a factory or city.



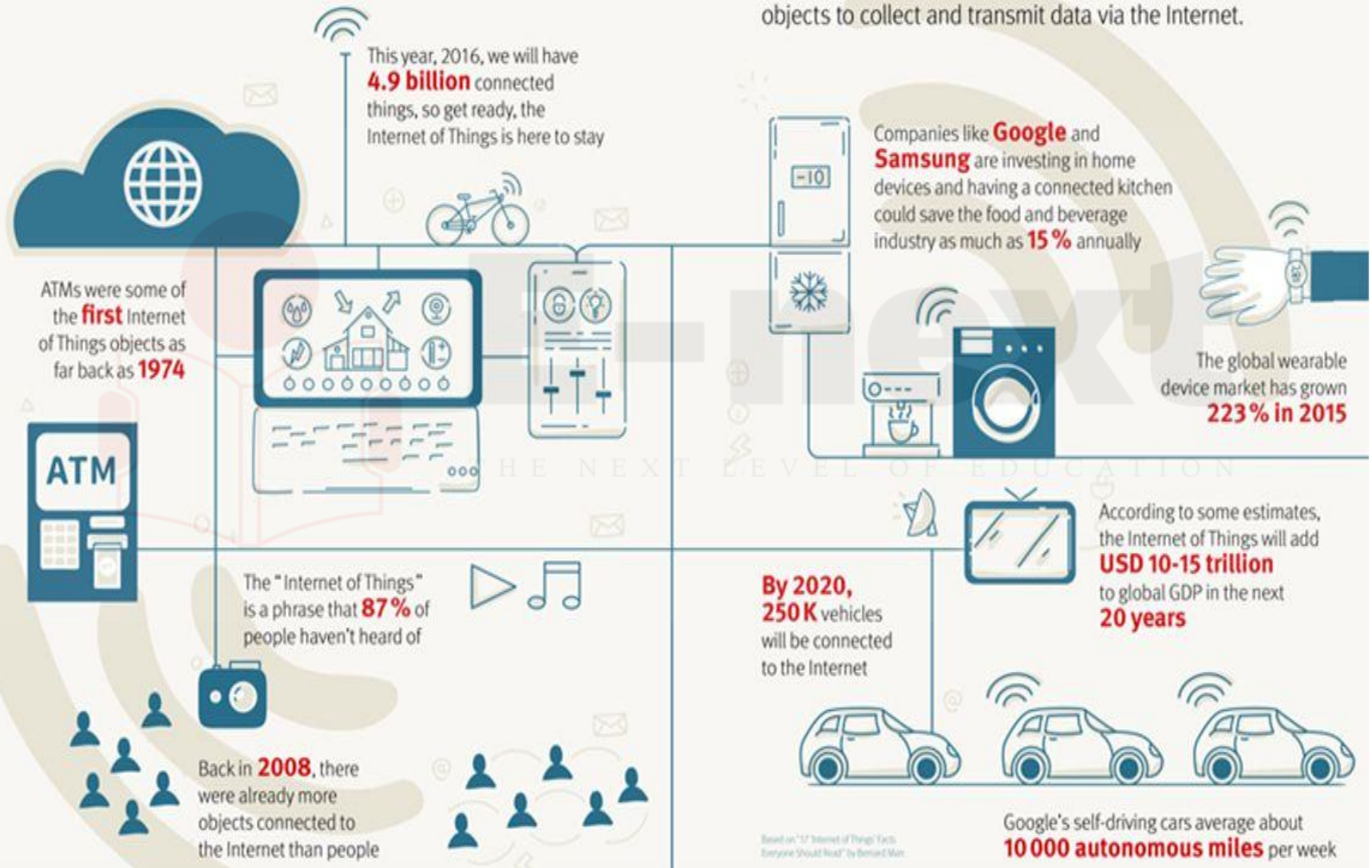
1. Monthly worldwide Google searches for the application 2. Monthly Tweets containing the application name and #IOT 3. Monthly LinkedIn Posts that include the application name. All metrics valid for Q4/2014.
Sources: Google, Twitter, LinkedIn, IoT Analytics

INTERNET OF THINGS Evolution

- the Internet has gone through roughly four generations of deployment culminating in the IoT:
- **1. Information technology (IT):** PCs, servers, routers, firewalls, and so on, bought as IT devices by enterprise IT people, primarily using wired connectivity.
- **2. Operational technology (OT):** Machines/appliances with embedded IT built by non-IT companies, such as medical machinery, SCADA (supervisory control and data acquisition), process control, and kiosks bought as appliances by enterprise OT people and primarily using wired connectivity.
- **3. Personal technology:** Smartphones, tablets, and ebook readers bought as IT devices by consumers (employees) exclusively using wireless connectivity and often multiple forms of wireless connectivity.
- **4. Sensor/actuator technology:** Single-purpose devices bought by consumers, IT, and OT people exclusively using wireless connectivity, generally of a single form, as part of larger systems.

The Internet of Things a very short story

The Internet of Things is the network of physical devices, vehicles, buildings and so on embedded with electronics, software, sensors and network connectivity that enable these objects to collect and transmit data via the Internet.



Layers of the Internet of Things

- **Some resources define it as 5 or 6 or 7 layer** , at all you will find these layers
- **Sensors and actuators:** These are the things. Sensors observe their environment and report back quantitative measurements of such variables as temperature, humidity, presence or absence of some observable,. Actuators operate on their environment, such as changing a thermostat setting or operating a valve.
- **Connectivity:** wireless or wired link into a network to send collected data to the appropriate data center (sensor) or receive operational commands from a controller site (actuator).
- **Capacity:** The network supporting the devices must be able to handle a potentially huge flow of data.
- **Storage:** There needs to be a large storage facility to store and maintain backups of all the collected data. This is typically a cloud

Internet of Things Reference Model

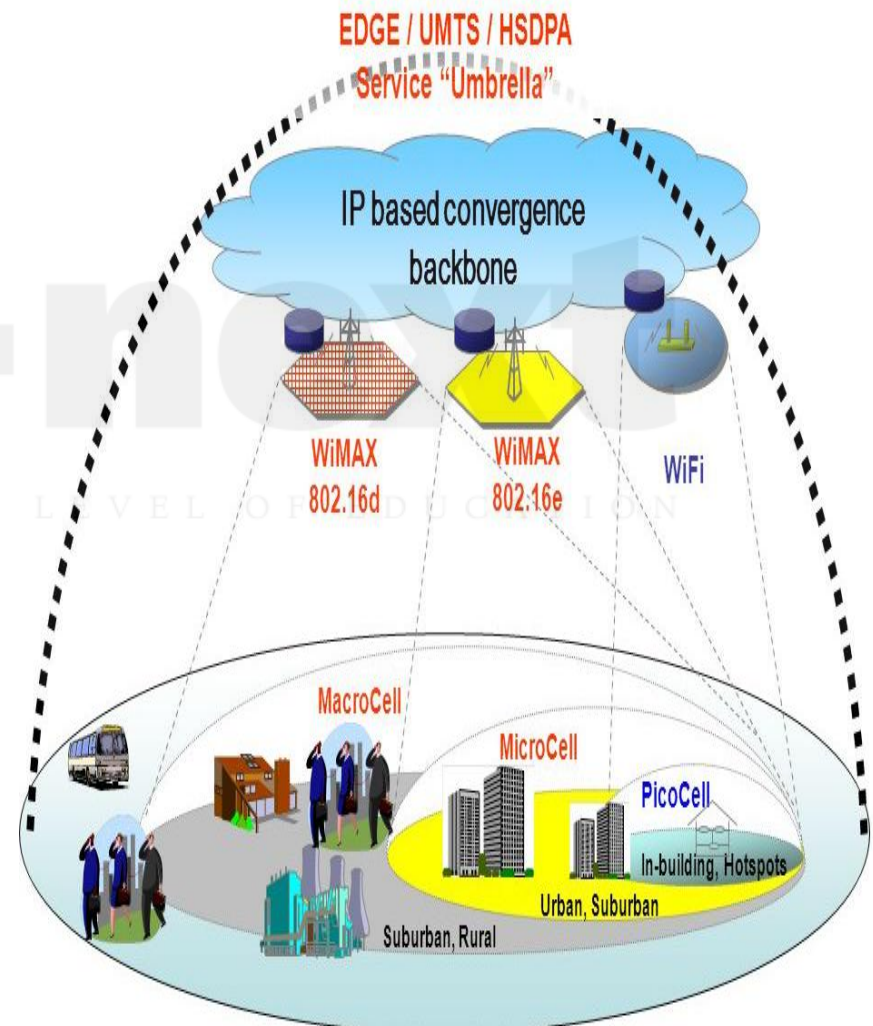
Levels



NETWORK CONVERGENCE

- It is The provision of telephone, video, and data communication services within a single network.
- the major attributes of the three-layer model of enterprise communications.
- convergence involves moving an organization's voice, video, and image traffic to a single network infrastructure.
- This often involves integrating distinct voice and data networks into a single network infrastructure and extending the infrastructure to support mobile users.
- The foundation of this convergence is packet-based transmission using the Internet Protocol (IP).
- Using IP packets to deliver all varieties of communications traffic, sometimes referred to as everything over IP, enables the underlying infrastructure to deliver a wide range of useful applications to business users.
- Convergence brings many benefits, including simplified network management, increased efficiency, and greater flexibility at the application level.
- a converged network infrastructure provides a predictable platform on which to build new add applications that combine video, data, and voice.

NETWORK CONVERGENCE



NETWORK CONVERGENCE

- You can think of this convergence in terms of a three-layer model of enterprise communications:
- **Application convergence:**
- These are seen by the end users of a business.
- Convergence integrates communications applications, such as voice calling (telephone), voice mail, e-mail, and instant messaging, with business applications, such as workgroup collaboration, customer relationship management, and backoffice functions.
- With convergence, applications provide rich features that incorporate voice, data, and video in a seamless, organized, and value-added manner.
- One example is multimedia messaging, which enables a user to use a single interface to access messages from a variety of sources (for example, office



NETWORK CONVERGENCE

- It is The provision of telephone, video, and data communication services within a single network.
- You can think of this convergence in terms of a three-layer model of enterprise communications:
- **Enterprise services:**
- At this level, the manager deals with the information network in terms of the services that must be available to ensure that users can take full advantage of the applications that they use.
- For example, network managers need to make sure that appropriate privacy mechanisms and authentication services are in place to support convergence based applications.
- They may also be able to track user locations to support remote print services and network storage facilities for mobile workers.



NETWORK CONVERGENCE:

Infrastructure:

- The network and communications infrastructure consists of the communication links, LANs, WANs, and Internet connections available to the enterprise. Increasingly, enterprise network
- infrastructure also includes private/public cloud connections to data centers that host high-volume data
- storage and web services.
- A key aspect of convergence at this level is the ability to carry voice, image, and
- video over networks that were originally designed to carry data traffic. Infrastructure convergence has also
- occurred for networks that were designed for voice traffic. For example, video, image, text, and data are
- routinely delivered to smartphone users over cell phone networks

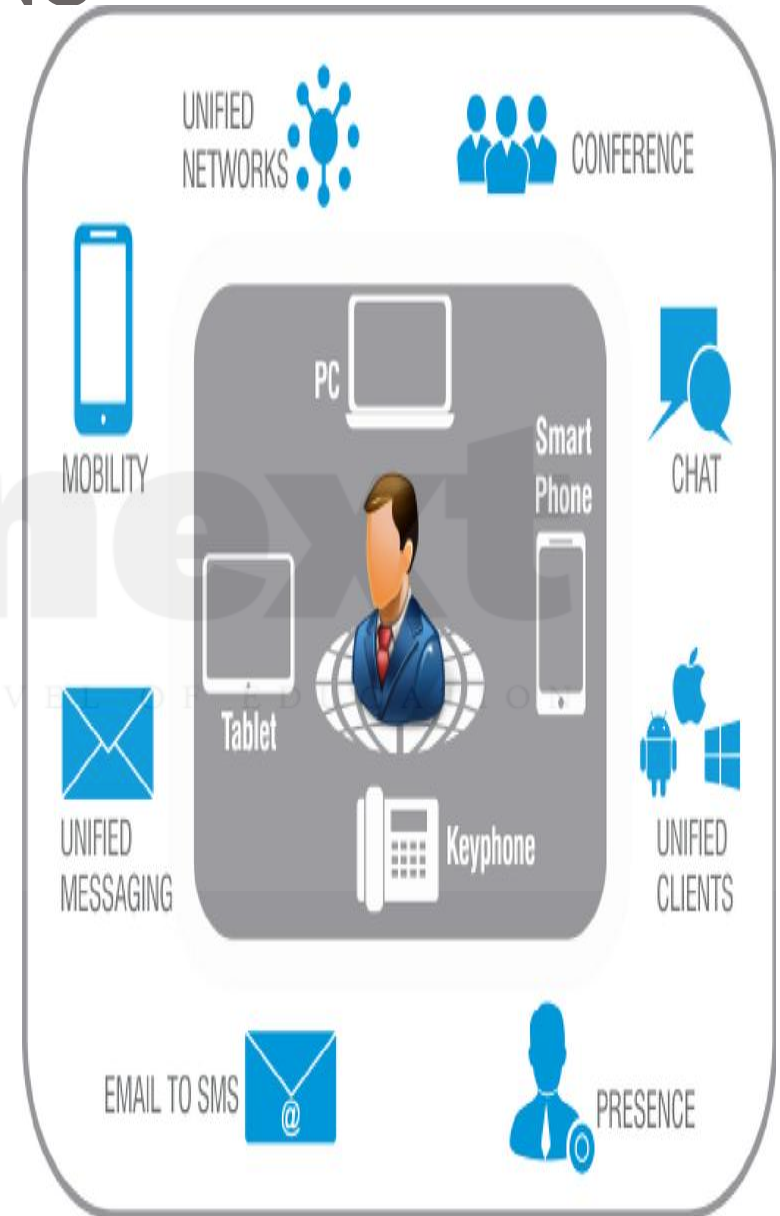


NETWORK CONVERGENCE

- **The following list summarizes three key benefits of IP network convergence:**
- **Cost savings:** provide significant double-digit percent reductions in network administration, maintenance, and operating costs; converging legacy networks onto a single IP network enables better use of existing resources, and implementation of centralized capacity planning, asset management, and policy management.
- **Effectiveness:** The converged environment has the potential to provide users with great flexibility, irrespective of where they are. IP convergence allows companies to create a more mobile workforce. Mobile workers can use a virtual private network (VPN) to remotely access business applications and communication services on the corporate network.
- **Transformation:** enables the enterprise-wide adoption of global standards and best practices, thus providing better data, enhanced real-time decision making, and improved execution of key business processes and operations.. Given their reliance on ICT (information and communications technology), today's converged

UNIFIED COMMUNICATIONS

- A concept related to network convergence is unified communications (UC).
- It is The integration of real-time enterprise communication services such as instant messaging, presence information, voice (including IP telephony), web and video conferencing, and speech recognition with nonreal- time communication services such as unified messaging (integrated voice mail, e-mail, SMS, and fax).
- Whereas enterprise network convergence focuses on the consolidation of traditionally distinct voice, video, and data communications networks into a common infrastructure, UC focuses on the integration of real-time communication services to optimize business processes.
- As with converged enterprise networks, IP is the cornerstone on which UC systems are built.



UNIFIED COMMUNICATIONS:

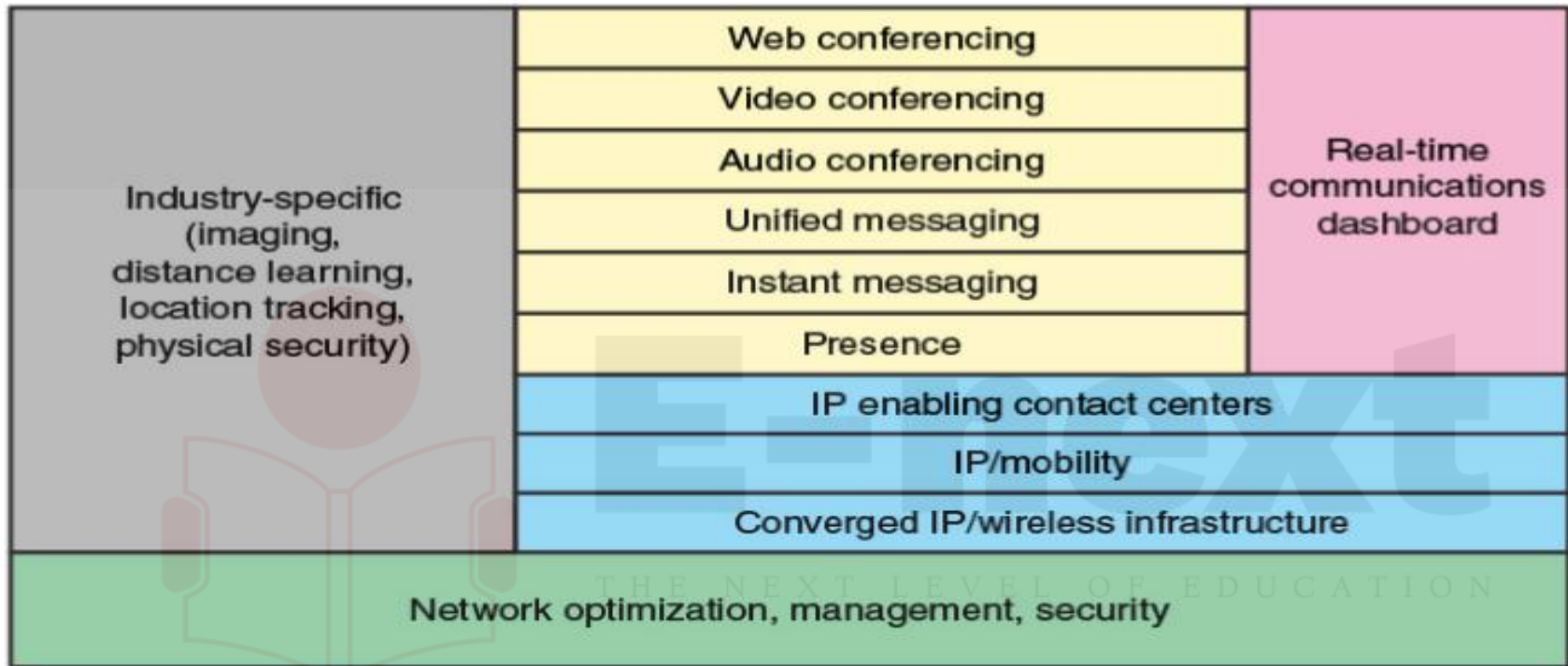


FIGURE 1.9 Elements of a Unified Communications Architecture

- Key elements of UC include the following:
- UC systems typically provide a unified user interface and consistent user experience across multiple device and media.
- UC merges real-time communications services with non-real-time services and business process applications.

UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows:
- **Realtime communications (RTC) dashboard:**
- An RTC dashboard is a key component of UC architecture.
- This is the element that provides UC users with a unified user interface across communication devices.
- Ideally, the user has a consistent interface no matter what communication device the user is currently using, whether it is a cell phone, desktop system, or office telephone attached to the corporate private branch exchange (PBX).
- dashboards provide access to realtime communication services such as instant messaging, audio and video conferencing, and interactive whiteboards; RTC dashboards also provide access to non-real-time services such as unified messaging (e-mail, voice mail, fax, and SMS) in unified view.



UNIFIED COMMUNICATIONS

- **The key elements of this architecture are as follows:**
- **Web conferencing:**
- Refers to live meetings or presentations in which participants access the meeting or presentation via a mobile device or the web, either over the Internet, or corporate intranet.

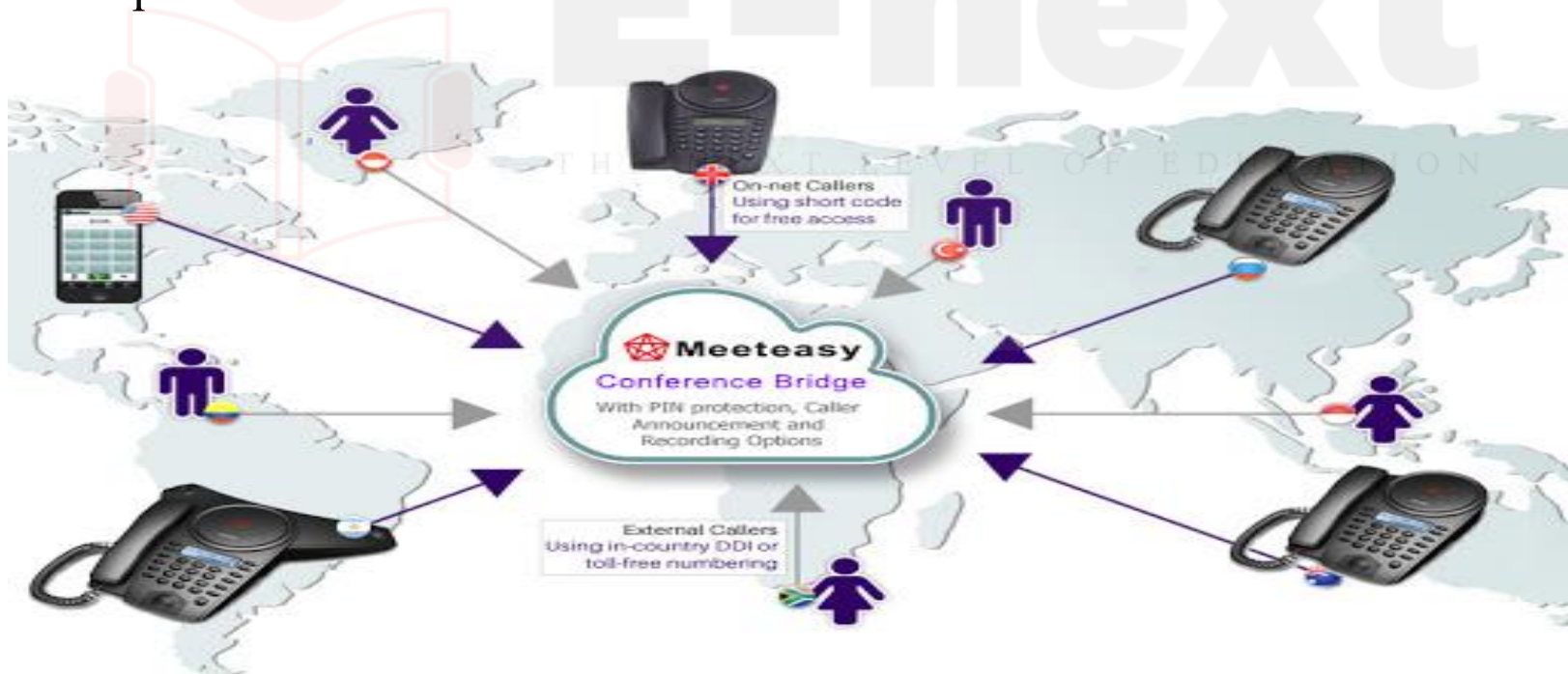


UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows:

Audio conferencing:

- Also called conference calling, refers to a live meeting in which participants are linked together for audio transmission and reception.
- A participant may be on a landline, mobile phone, or at a “softphone” a computer

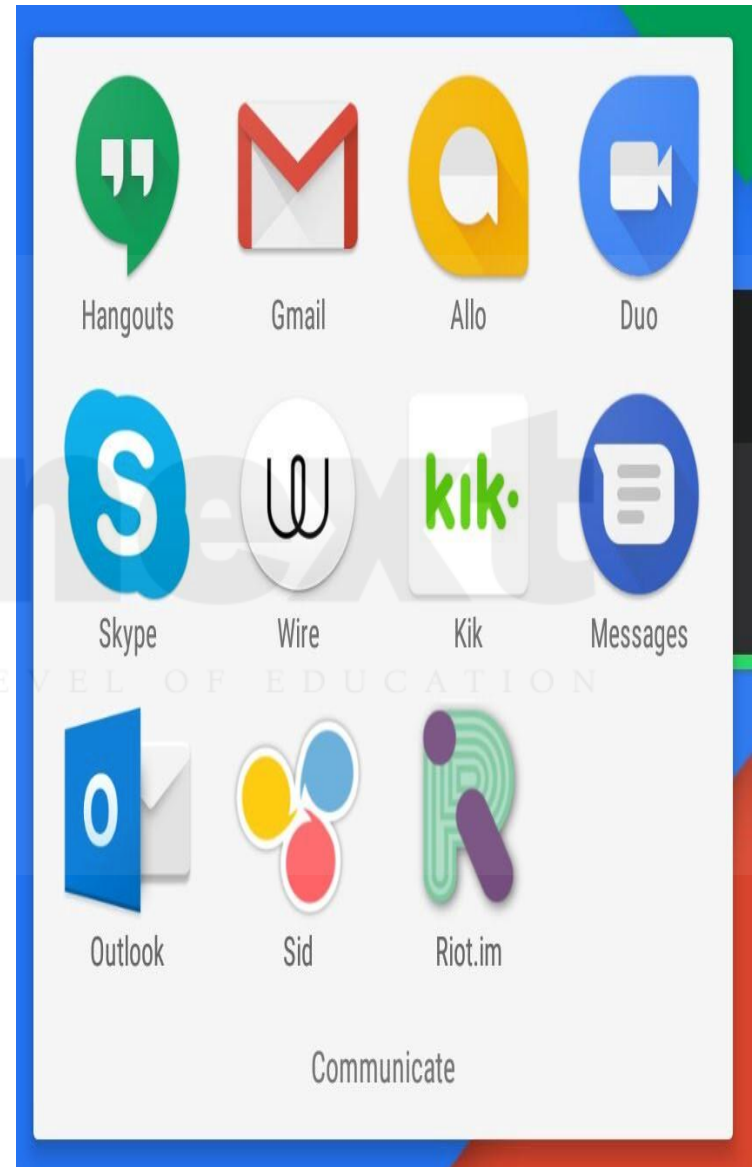


UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows:

Unified messaging:

- Unified messaging systems provide a common repository for messages from multiple sources.
- It allows users to retrieve saved e-mail, voice mail, and fax messages from a computer, telephone, or mobile device
- Computer users can select and play voice-mail recordings that appear in their unified messaging inboxes.
- Telephone users can both retrieve voice mail and hear text-to-voice translations of e-mail messages.
- Messages of any type can be saved, answered, filed, sorted, and forwarded. Unified messaging systems relieve business users from having to monitor multiple voice mailboxes by enabling voicemail messages received by both office phones and cell phones to be saved to the same mailbox.



UNIFIED COMMUNICATIONS

- **The key elements of this architecture are as follows:**
- **Instant messaging (IM):**
- Real-time text-based messaging between two or more participants.
- IM is similar to online chat because it is text-based and exchanged bidirectionally in real time.
- IM is distinct from chat in that IM clients use contact (or buddy) lists to facilitate connections between known users, whereas online chat can include text based exchanges between anonymous users.



UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows:
- Video teleconferencing (VTC):
- Videoconferencing allows users in two or more locations to interact simultaneously via two-way video and audio transmission
- UC systems enable users to participate in video conferences via **desktop** computers, smartphones, and mobile devices.



UNIFIED COMMUNICATIONS

- **The key elements of this architecture are as follows:**

Presence:

- The capability to determine, in real time, where someone is, how that person prefers to be reached, and even what the person is currently doing.
- Presence information shows the individual's availability state before co-workers attempt to contact them person.
- It was once considered simply an underlying technology to instant messaging (for example, “available to chat” or “busy”) but has been broadened to include whether co-workers are currently on office or mobile phones, logged in to a computer, involved in a video call or in a meeting, or out of the office for lunch or vacation.
- A co-worker's geographic location is becoming more common as an element in presence information for a number of business reasons, including the capability to quickly respond to customer emergencies.
- Business has embraced presence information because it facilitates more efficient and effective



UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows:

IP enabling contact centers:

- Refers to the use of IP-based unified communications to enhance customer contact center functionality and performance.
- The unified communications infrastructure makes use of presence technology to enable customers and internal enterprise employees to be quickly connected to the required expert or support person.
- In addition, this technology supports mobility, so that call center personnel need not be located at a particular office or remain in a particular place.
- Finally, the UC infrastructure enables the call center employee to quickly access other employees and information assets, including data, video, image, and audio.



UNIFIED COMMUNICATIONS

- **The key elements of this architecture are as follows:**
IP/mobility:
- Refers to the delivery of information to and collection of information from enterprise personnel who are usually mobile, using an IP network infrastructure.
- In a typical enterprise, upward of 30 percent of employees use some form of weekly remote access technology in the performance of their jobs.



UNIFIED COMMUNICATIONS

- The key elements of this architecture are as follows: **Converged IP/wireless infrastructure:**
- A unified networking and communications-based IP packet transfer to support voice, data, and video transmission and can be extended to include local and wide-area wireless communications
- UC-enabled mobile devices are able to switch between Wi-Fi and cellular systems in the middle of a communication session.
- For example, a UC user could receive a co-worker's call via a smartphone connected to Wi-Fi network at home, continue the conversation while driving to work over a cellular network connection, and could end the call at the office while connected to the business's Wi-Fi network.
- Both handoffs (home Wi-Fi to cellular and cellular to office Wi-Fi) would take place seamlessly and transparently without dropping the call.



UNIFIED COMMUNICATIONS

- The importance of UC is not only that it integrates communication channels but also that it offers a way to integrate communication functions and business applications. Three major categories of benefits are typically realized by organizations that use UC:
- **Personal productivity gains:**
- Presence information helps employees find each other and choose the most effective way to communicate in real time.
- Less time is wasted calling multiple numbers to locate co-workers or checking multiple worked-related voice mailboxes.
- Calls from VIP contacts can be routed simultaneously to all of a UC user's phone devices (office phone, softphone, smartphone, home phone) to ensure faster responsiveness to customers, partners, and co-workers.
- With mobile presence information capabilities, employees who are



UNIFIED COMMUNICATIONS

- The importance of UC is not only that it integrates communication channels but also that it offers a way to integrate communication functions and business applications. Three major categories of benefits are typically realized by organizations that use UC:
- **Workgroup performance gains:**
- UC systems support real-time collaboration among team members, which facilitates workgroup performance improvements.
- Examples include the use of presence information to speed identification of an available individual with the right skills a work team needs to address a problem.
- Enhanced conferencing capabilities with desktop VTC and interactive white boards and automated business rules to route or escalate communications also help to increase workgroup performance.



UNIFIED COMMUNICATIONS

- The importance of UC is not only that it integrates communication channels but also that it offers a way to integrate communication functions and business applications. Three major categories of benefits are typically realized by organizations that use UC:
- **Enterprise level process improvements:**
- IP convergence enables UC to be integrated with enterprise wide and departmental-level applications, business processes, and workflows.
- UC-enabled enhanced communications with customers, suppliers, and business partners are redefining best practices for customer relationship management (CRM), supply chain management (SCM), and other enterprise-wide applications and are transforming relationships among members of business networks.

